Supplementary Material

SignMuon, MuonSign, and the Role of Error Feedback

A Appendix

A.1 Preliminaries: LMO, Muon, and generalized smoothness

This subsection states in full the definitions that Section 3 uses in abbreviated form.

A *Linear Minimization Oracle* (LMO) is an operator $A(\cdot)$ that provides the solution to a linear minimization problem over a feasible set. In particular, the constraints can be defined via the unit ball of a given norm:

$$A(\mathbf{X}) = \arg\min_{\mathbf{Y}\in\{\mathbf{Y}\in\mathcal{S}\mid\|\mathbf{Y}\|\leq 1\}} \langle \mathbf{X}, \mathbf{Y}\rangle, \tag{1}$$

where $\mathbf{X} \in \mathbb{R}^{m\times n}$ is the input matrix, $\mathcal{S} \subset \mathbb{R}^{m\times n}$ is a convex set, and $\langle \mathbf{X}, \mathbf{Y}\rangle = \sum_{i,j} X_{ij}Y_{ij}$ is the standard matrix inner product (Pethick et al. 2025). Such an operator selects the direction of steepest descent with respect to the specified norm. Its analytical and practical role is discussed in detail in the literature on norm-constrained LMO optimizers (Pethick et al. 2025; Kovalev 2025; Riabinin et al. 2025; Bernstein and Newhouse 2024; Bernstein 2025), and the catalogue of usable geometries keeps growing (Cesista 2025; Kravatskiy et al. 2025).

Muon is the instance in which each matrix layer carries the spectral norm, so that the oracle uses the matrix structure of the gradient and the update direction is obtained by orthogonalizing the gradient matrix (Bernstein and Newhouse 2024). Let $\mathbf{M} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ be the singular value decomposition (SVD) of the matrix $\mathbf{M} \in \mathbb{R}^{m\times n}$, where $\mathbf{U} \in \mathbb{R}^{m\times r}$ and $\mathbf{V} \in \mathbb{R}^{n\times r}$ are the orthonormal matrices of singular vectors, $\mathbf{\Sigma} \in \mathbb{R}^{r\times r}$ is the diagonal matrix of singular values, and $r = \operatorname{rank}(\mathbf{M})$. Then the LMO direction is $A(\mathbf{M}) = -\mathbf{U}\mathbf{V}^\top$: Muon selects an orthonormal update direction corresponding to the solution of the linear minimization problem induced by the spectral norm geometry. Such orthogonalization produces more isotropic updates and prevents a small number of dominant singular directions from controlling the optimization process, which is particularly beneficial for training DNN (Bernstein and Newhouse 2024; Essential AI 2025). In the original Muon implementation the computation of $\mathbf{U}_t\mathbf{V}_t^\top$ is approximated using Newton-Schulz iterations and related polynomial schemes, which enable efficient matrix orthogonalization without an explicit SVD computation (Amsel et al. 2026; Li and Hong 2025; Liu et al. 2025); Algorithm A1 gives the procedure and the coefficients we use.

Exact zeros and the sign convention. Zeros in the compressed object are not exotic: a unit that is inactive across a whole local batch zeroes a row of the momentum, and that row survives both the LMO and the sign. Resolving $\operatorname{sign}(0)$ to 0 would therefore leave a third symbol in the alphabet carrying non-negligible mass, making the channel wider than one bit and realizable only with an entropy coder. Appendix A.14 quantifies the excess. We resolve it to an independent random ± 1 instead, on every channel in this paper. A zeroed coordinate carries no directional information, so this does not degrade the expected descent, and it makes exact two statements the ternary convention only approximates: $\|\mathbf{s}_t\|_F = \sqrt{mn}$, which the unit-gain rule (9) assumes, and the contraction identity of Lemma 4.

Finally, Corollary 2 needs Assumption 2 only in its weaker layer-wise (L^0, L^1) form, which replaces the constant L_i^g by $L_i^{0,g} + L_i^{1,g}\|\nabla_i g(\mathbf{X})\|_*$ (Riabinin et al. 2025; Pethick et al. 2026): for every layer i and all $\mathbf{X}, \mathbf{Y}$,

$$\|\nabla_i g(\mathbf{X}) - \nabla_i g(\mathbf{Y})\|_* \leq \left(L_i^{0,g} + L_i^{1,g}\|\nabla_i g(\mathbf{X})\|_*\right)\|\mathbf{X}_i - \mathbf{Y}_i\|_{2\to 2}, \tag{2}$$

again with one pair of constants per layer for $g = f$ and per layer and client for $g = f_j$.

A.2 Divergence on linear objectives: reduction and momentum

The reduction quoted in Section 4 collapses each of the three methods to a single scalar inequality and removes momentum from the discussion entirely.

Proposition 1 (Reduction on linear objectives) *Run any of the three methods* (10) *on the linear objective* (11) *with* $\mathbf{G} \neq \mathbf{0}$*, from an arbitrary* $\mathbf{X}_0$*, with any momentum coefficient* $\mu \in [0, 1)$ *under either the Standard or the Nesterov rule. Then* $\tilde{\mathbf{M}}_t = \gamma_t\,\mathbf{G}$ *with* $\gamma_t > 0$*, the update direction is the constant matrix* $\mathbf{s}(\mathbf{G})$ *obtained by substituting* $\mathbf{G}$ *for* $\tilde{\mathbf{M}}_t$ *in* (10)*, and*

$$f(\mathbf{X}_{t+1}) - f(\mathbf{X}_t) = -\eta_t\left\langle \mathbf{G},\, \mathbf{s}(\mathbf{G})\right\rangle.$$

In particular, if $\langle \mathbf{G}, \mathbf{s}(\mathbf{G})\rangle < 0$ *then* f *strictly increases at every iteration for any* $\eta_t > 0$*, and* $f(\mathbf{X}_t) \to +\infty$ *whenever* $\sum_t \eta_t = \infty$ *(e.g. for any constant step size).*

Proof of Proposition 1. On the linear objective (11) the gradient is globally constant, $\mathbf{G}_t = \mathbf{G}$, so the momentum buffer of (7) is $\mathbf{M}_t = \sum_{i=0}^{t-1} \mu^i \, \mathbf{G} = \frac{1-\mu^t}{1-\mu} \, \mathbf{G}$. Both momentum rules then return a positive multiple of $\mathbf{G}$,

$$\tilde{\mathbf{M}}_t = \gamma_t \, \mathbf{G}, \qquad \gamma_t = \begin{cases} \frac{1-\mu^t}{1-\mu}, & \text{(Standard)}, \\ 1 + \mu \, \frac{1-\mu^t}{1-\mu}, & \text{(Nesterov)}, \end{cases} \tag{3}$$

because $\mu \in [0, 1)$. The elementwise $\mathrm{sign}(\cdot)$ and the Muon LMO $\mathrm{polar}(\cdot)$ are each invariant under multiplication by a positive scalar, so evaluating (10) at $\tilde{\mathbf{M}}_t = \gamma_t \mathbf{G}$ returns the same matrix as evaluating it at $\mathbf{G}$; that is, $\mathbf{s}_t = \mathbf{s}(\mathbf{G})$ for every t, independently of μ and of the momentum variant. Hence $f(\mathbf{X}_{t+1}) - f(\mathbf{X}_t) = \langle \mathbf{G}, \mathbf{X}_{t+1} - \mathbf{X}_t \rangle = -\eta_t \langle \mathbf{G}, \mathbf{s}(\mathbf{G}) \rangle$, which is (12). If $\langle \mathbf{G}, \mathbf{s}(\mathbf{G}) \rangle < 0$, then $f(\mathbf{X}_t) = f(\mathbf{X}_0) - \langle \mathbf{G}, \mathbf{s}(\mathbf{G}) \rangle \sum_{i=0}^{t-1} \eta_i$ increases strictly at every step and diverges to $+\infty$ whenever $\sum_t \eta_t = \infty$. ■

By Proposition 1, each divergence theorem reduces to exhibiting a single gradient $\mathbf{G}$ with $\langle \mathbf{G}, \mathbf{s}(\mathbf{G}) \rangle < 0$; the three proofs below do exactly this, and momentum needs no further comment.

A.3 Proof of Theorem 1 (Divergence of SignMuon)

Theorem 1 (Divergence of SignMuon) *There is a matrix $\mathbf{G} \in \mathbb{R}^{4\times 4}$ with $\big\langle \mathbf{G}, \mathrm{sign}(\mathrm{polar}(\mathbf{G})) \big\rangle = -\frac{42468}{103} < 0$. Hence SignMuon ascends on* (11)*: f strictly increases at every iteration for all $\eta_t > 0$, all $\mu \in [0, 1)$, and both momentum variants.*

For SignMuon $\mathbf{s}(\mathbf{G}) = \mathrm{sign}(\mathrm{polar}(\mathbf{G}))$, so by (12) it suffices to construct $\mathbf{G} \in \mathbb{R}^{4\times 4}$ with $\langle \mathbf{G}, \mathrm{sign}(\mathrm{polar}(\mathbf{G})) \rangle < 0$. The size is not arbitrary. Writing the polar decomposition $\mathbf{G} = \mathbf{Q}\mathbf{H}$ with $\mathbf{H} \succeq 0$, and using that $\mathrm{polar}(\mathbf{G}) = \mathbf{Q}$ and that $\mathbf{H}$ is symmetric,

$$\langle \mathbf{G}, \mathrm{sign}(\mathrm{polar}(\mathbf{G})) \rangle = \big\langle \mathbf{H}, \mathrm{sym}(\mathbf{Q}^\top \mathrm{sign}(\mathbf{Q})) \big\rangle, \tag{4}$$

so a counterexample with polar factor $\mathbf{Q}$ exists precisely when $\lambda_{\min}\big(\mathrm{sym}(\mathbf{Q}^\top \mathrm{sign}(\mathbf{Q}))\big) < 0$, a question about $O(n)$ alone. For $n = 2$ this settles it: every $\mathbf{Q} \in O(2)$ is $\left(\begin{smallmatrix} c & -s \\ s & c \end{smallmatrix}\right)$ or $\left(\begin{smallmatrix} c & s \\ s & -c \end{smallmatrix}\right)$, and in both cases $\mathbf{Q}^\top \mathrm{sign}(\mathbf{Q}) = (|c| + |s|)\mathbf{I} +$ (antisymmetric), whose symmetric part is positive definite, so $\langle \mathbf{G}, \mathrm{sign}(\mathrm{polar}(\mathbf{G})) \rangle = (|c| + |s|) \,\mathrm{tr}\, \mathbf{H} > 0$ and SignMuon cannot ascend. At $n = 3$ several independent searches over $O(3)$ found no negative eigenvalue; at $n = 4$ they do, and the $\mathbf{O}$ below is one such instance (its spectrum is $(-0.4175, 1.5825, 2, 2)$, so it is not even the worst).

We construct $\mathbf{G} \in \mathbb{R}^{4\times 4}$ by defining a specific orthogonal matrix $\mathbf{O}$ and a specific rank-1 principal component $\mathbf{u}_1 \mathbf{v}_1^\top$. Let the orthogonal matrix $\mathbf{O}$ be given by the following exact rational numbers:

$$\mathbf{O} = \frac{1}{103} \begin{pmatrix} 101 & 20 & 2 & -2 \\ -20 & 97 & 20 & -20 \\ -2 & 20 & 2 & 101 \\ -2 & 20 & -101 & -2 \end{pmatrix}. \tag{5}$$

Because no entry is zero, its element-wise sign matrix $\mathbf{S} = \mathrm{sign}(\mathbf{O})$ is uniquely defined. Now, let $\mathbf{u}_1$ and $\mathbf{v}_1$ be the following exact unit vectors:

$$\mathbf{u}_1 = \frac{1}{\sqrt{309}} \begin{pmatrix} 10 \\ -3 \\ 10 \\ 10 \end{pmatrix}, \quad \mathbf{v}_1 = \frac{1}{\sqrt{309}} \begin{pmatrix} 10 \\ 3 \\ -10 \\ 10 \end{pmatrix}. \tag{6}$$

One can easily verify that $\mathbf{O}\mathbf{v}_1 = \mathbf{u}_1$, meaning $\mathbf{u}_1$ and $\mathbf{v}_1$ act perfectly as left and right singular vectors for this orthogonal space. The crucial feature of this geometry is that the Frobenius inner product between this rank-1 component and the sign matrix $\mathbf{S}$ yields an exact, strictly negative fraction:

$$\langle \mathbf{u}_1 \mathbf{v}_1^\top, \mathbf{S} \rangle = \mathbf{u}_1^\top \mathbf{S} \mathbf{v}_1 = -\frac{43}{103}. \tag{7}$$

We construct the gradient matrix $\mathbf{G}$ by assigning a large singular value ($\sigma_1 = 1001$) to this pathological component and a singular value of 1 to the rest of the orthogonal space:

$$\mathbf{G} \approx \begin{pmatrix} 324.6 & 97.3 & -323.6 & 323.6 \\ -97.3 & -28.2 & 97.3 & -97.3 \\ 323.6 & 97.3 & -323.6 & 324.6 \\ 323.6 & 97.3 & -324.6 & 323.6 \end{pmatrix}. \tag{8}$$

Indeed, since $\mathbf{G} = 1000 \, \mathbf{u}_1 \mathbf{v}_1^\top + \mathbf{O}$ and $\mathbf{O}\mathbf{v}_1 = \mathbf{u}_1$,

$$\begin{aligned} \mathbf{G}\mathbf{v}_1 &= 1000 \, \mathbf{u}_1 (\mathbf{v}_1^\top \mathbf{v}_1) + \mathbf{O}\mathbf{v}_1 \\ &= 1001 \, \mathbf{u}_1, \end{aligned}$$

so $(\mathbf{u}_1, \mathbf{v}_1)$ is a singular pair of $\mathbf{G}$ with $\sigma_1 = 1001$. For any $\mathbf{w} \perp \mathbf{v}_1$ we have $\mathbf{G}\mathbf{w} = \mathbf{O}\mathbf{w}$; as $\mathbf{O}$ is orthogonal and $\mathbf{O}\mathbf{v}_1 = \mathbf{u}_1$, the restriction $\mathbf{O}|_{\mathbf{v}_1^\perp}$ is an isometry onto $\mathbf{u}_1^\perp$, hence $\sigma_2 = \sigma_3 = \sigma_4 = 1$. Moreover, with $\mathbf{u}_k = \mathbf{O}\mathbf{v}_k$ for $k \geq 2$,

$$\textstyle\sum_{k\geq 2} \mathbf{u}_k \mathbf{v}_k^\top = \mathbf{O}\big(\mathbf{I} - \mathbf{v}_1\mathbf{v}_1^\top\big) = \mathbf{O} - \mathbf{u}_1\mathbf{v}_1^\top,$$

so $\mathbf{U}_G\mathbf{V}_G^\top = \mathbf{u}_1\mathbf{v}_1^\top + (\mathbf{O} - \mathbf{u}_1\mathbf{v}_1^\top) = \mathbf{O}$.

Substituting the resulting matrix $\mathbf{G}$ into the descent condition yields:

$$\langle \mathbf{G}, \mathbf{S}\rangle = \langle 1000\,\mathbf{u}_1\mathbf{v}_1^\top + \mathbf{O}, \mathbf{S}\rangle = 1000\langle \mathbf{u}_1\mathbf{v}_1^\top, \mathbf{S}\rangle + \langle \mathbf{O}, \mathbf{S}\rangle. \tag{9}$$

The inner product $\langle \mathbf{O}, \mathbf{S}\rangle$ is equivalent to the L_1 norm (sum of absolute values) of $\mathbf{O}$, which equals exactly $\frac{532}{103}$. Therefore:

$$\langle \mathbf{G}, \mathbf{S}\rangle = 1000\left(-\frac{43}{103}\right) + \frac{532}{103} = -\frac{42468}{103} \approx -412.31. \tag{10}$$

Here $\mathbf{S} = \operatorname{sign}(\mathbf{O}) = \operatorname{sign}(\operatorname{polar}(\mathbf{G})) = \mathbf{s}(\mathbf{G})$, so $\langle \mathbf{G}, \mathbf{s}(\mathbf{G})\rangle = -\frac{42468}{103} < 0$. By Proposition 1, SignMuon strictly ascends, $f(\mathbf{X}_{t+1}) - f(\mathbf{X}_t) = \frac{42468}{103}\,\eta_t > 0$ at every iteration, for every $\eta_t > 0$, every $\mu \in [0, 1)$, and both momentum variants; $f(\mathbf{X}_t) \to +\infty$ under any non-summable step size. ■

A.4 Proof of Theorem 2 (Divergence of MuonUSign)

Theorem 2 (Divergence of MuonUSign) *There is a matrix $\mathbf{G} \in \mathbb{R}^{5\times 5}$ with $\big\langle \mathbf{G}, \operatorname{polar}(\operatorname{sign}(\mathbf{G}))\big\rangle \approx -13.89 < 0$. Hence MuonUSign ascends on* (11) *for all $\eta_t > 0$, all $\mu \in [0, 1)$, and both momentum variants.*

MuonUSign (Algorithm A4) applies the sign *before* the LMO, so $\mathbf{s}(\mathbf{G}) = \operatorname{polar}(\operatorname{sign}(\mathbf{G}))$. By (12) it suffices to construct $\mathbf{G} \in \mathbb{R}^{5\times 5}$ with $\langle \mathbf{G}, \operatorname{polar}(\operatorname{sign}(\mathbf{G}))\rangle < 0$.

The size is again not arbitrary, and here the question reduces to *sign patterns*. With $\mathbf{S} = \operatorname{sign}(\mathbf{G})$ and $\mathbf{D} = \operatorname{polar}(\mathbf{S})$, the fact that $\mathbf{G}$ vanishes off the support of $\mathbf{S}$ gives

$$\big\langle \mathbf{G}, \operatorname{polar}(\operatorname{sign}(\mathbf{G}))\big\rangle = \textstyle\sum_{i,j} |G_{ij}|\, S_{ij} D_{ij}, \tag{11}$$

and likewise $\langle \mathbf{G}, \operatorname{sign}(\mathbf{D})\rangle = \sum_{i,j} |G_{ij}|\, S_{ij} \operatorname{sign}(D_{ij})$ for the MuonSign step of Theorem 3. So an ascent instance exists at a given shape precisely when some sign pattern $\mathbf{S}$ there has a *mismatched* entry, $S_{ij}D_{ij} < 0$. Under our convention $\mathbf{S} \in \{\pm 1\}^{m\times n}$ always, and mismatch existence is invariant under row and column sign flips ($\operatorname{polar}(\mathbf{P}_1\mathbf{S}\mathbf{P}_2) = \mathbf{P}_1 \operatorname{polar}(\mathbf{S})\mathbf{P}_2$ for ± 1 diagonal $\mathbf{P}_1, \mathbf{P}_2$), so enumerating one representative per class settles the question outright (`code/counterexamples/enumerate_minimality.py`). It returns: *no* pattern at 4×4, at $3 \times n$ ($n \leq 9$) or at $2 \times n$ ($n \leq 12$) has a mismatch. Neither MuonUSign nor MuonSign can therefore ascend on any linear objective at those sizes at all. The first instances appear at 4×5 (720 of the 2^{12} classes) and, among square shapes, at 5×5, where the $\mathbf{S}$ below attains the deepest mismatch any 5×5 sign matrix admits, $|D_{4,2}| = 1/\sqrt{17}$. (Had we kept $\operatorname{sign}(0) = 0$, gradients with exact zeros would admit instances from 3×4 on; the convention removes them, along with their dependence on an entry being exactly zero.)

Fix the full-rank sign matrix $\mathbf{S} \in \{-1, 1\}^{5\times 5}$,

$$\mathbf{S} = \begin{pmatrix} -1 & -1 & 1 & 1 & 1 \\ -1 & -1 & 1 & -1 & -1 \\ 1 & -1 & 1 & 1 & -1 \\ 1 & 1 & -1 & -1 & 1 \\ 1 & 1 & 1 & -1 & 1 \end{pmatrix}, \tag{12}$$

and let $\mathbf{D} = \operatorname{polar}(\mathbf{S})$ be its (unique, since $\mathbf{S}$ is full rank) Muon LMO direction. A direct computation gives $D_{4,2} = -1/\sqrt{17} \approx -0.2425 < 0$ while $\operatorname{sign}(D_{i,j}) = S_{i,j}$ at all 24 other entries; equivalently, $\operatorname{sign}(\mathbf{D})$ and $\mathbf{S}$ disagree at exactly the single entry $(4, 2)$, where $S_{4,2} = +1$. We exploit this lone mismatch. Define

$$\mathbf{G} = \epsilon\,\mathbf{S} + (M - \epsilon)\,\mathbf{e}_4\mathbf{e}_2^\top, \qquad \epsilon > 0,\ M > \epsilon, \tag{13}$$

so that $G_{4,2} = M > 0$ and $G_{i,j} = \epsilon S_{i,j}$ otherwise; hence $\operatorname{sign}(\mathbf{G}) = \mathbf{S}$ and $\operatorname{polar}(\operatorname{sign}(\mathbf{G})) = \mathbf{D}$ for every $M > 0$. The descent inner product splits as

$$\langle \mathbf{G}, \mathbf{D}\rangle = \epsilon \sum_{(i,j)\neq(4,2)} S_{i,j}D_{i,j} \;+\; M\,D_{4,2} = \epsilon\,C + M\,D_{4,2}, \tag{14}$$

where $C := \sum_{(i,j)\neq(4,2)} |D_{i,j}| = 10.366 > 0$ is fixed (every such entry agrees in sign with $\mathbf{S}$), while $M\,D_{4,2} = -M/\sqrt{17} \to -\infty$. Thus $\langle \mathbf{G}, \mathbf{D}\rangle < 0$ for any $M > \sqrt{17}\,C\epsilon \approx 42.7\,\epsilon$; with $\epsilon = 1$, $M = 100$ one obtains $\langle \mathbf{G}, \mathbf{D}\rangle = -13.89$, for the exact polar factor that the theorem is stated over. By Proposition 1, MuonUSign strictly ascends on $f(\mathbf{W}) = \langle \mathbf{G}, \mathbf{W}\rangle$ for every $\eta_t > 0$, every $\mu \in [0, 1)$, and both momentum variants; $f(\mathbf{X}_t) \to +\infty$ under any non-summable step size. ■

A.5 Proof of Theorem 3 (Divergence of MuonSign)

Theorem 3 (Divergence of MuonSign) *For the* same *matrix* $\mathbf{G} \in \mathbb{R}^{5\times 5}$ *as in Theorem 2,* $\big\langle \mathbf{G}, \operatorname{sign}(\operatorname{polar}(\operatorname{sign}(\mathbf{G})))\big\rangle = -76 < 0$. *Hence MuonSign ascends on* (11) *for all* $\eta_t > 0$*, all* $\mu \in [0, 1)$*, and both momentum variants.*

MuonSign (Algorithm A5) signs the LMO output as well, so $\mathbf{s}(\mathbf{G}) = \operatorname{sign}(\operatorname{polar}(\operatorname{sign}(\mathbf{G}))) = \operatorname{sign}(\mathbf{D})$. We reuse the *same* $\mathbf{S}$ and $\mathbf{G}$ of (12)–(13): since $\operatorname{sign}(\mathbf{G}) = \mathbf{S}$, the bidirectional step is the constant matrix $\operatorname{sign}(\mathbf{D})$, which agrees with $\mathbf{S}$ at all 24 entries except $(4, 2)$, where $\operatorname{sign}(D_{4,2}) = -1 = -S_{4,2}$. Using $S^2_{i,j} = 1$ everywhere,

$$\begin{aligned}\langle \mathbf{G}, \operatorname{sign}(\mathbf{D})\rangle &= \epsilon \sum_{(i,j)\neq(4,2)} S_{i,j} \operatorname{sign}(D_{i,j}) + M \, \operatorname{sign}(D_{4,2}) \\ &= 24\,\epsilon - M.\end{aligned} \tag{15}$$

This is negative for every $M > 24\epsilon$; with $\epsilon = 1,\ M = 100$ it equals exactly -76. By Proposition 1, MuonSign strictly ascends on $f(\mathbf{W}) = \langle \mathbf{G}, \mathbf{W}\rangle$ for every $\eta_t > 0$, every $\mu \in [0, 1)$, and both momentum variants; $f(\mathbf{X}_t) \to +\infty$ under any non-summable step size. In particular, the *same* 5×5 linear instance defeats both the uplink-only (MuonUSign) and the bidirectional (MuonSign) placements. ■

A.6 Comparison with the convergence claim for SignMuon

Concurrently, Mishra, Trivedi, and Kumar (2026) introduced the same sign-after-LMO method under the same name *SignMuon*, with update direction $\mathbf{S}_t = \operatorname{sign}(\operatorname{polar}(\mathbf{M}_t))$ rescaled to $\mathbf{D}_t = \mathbf{S}_t/\sqrt{mn}$; this coincides with the SignMuon step of Section 4 after absorbing $1/\sqrt{mn}$ into η. For this method they establish an $\mathcal{O}(1/\sqrt{T})$ stationarity guarantee. We show that this guarantee does not contradict Theorem 1: the finalized rate is proved for a different update direction, and for $\mathbf{D}_t$ their bound is uninformative on the instance of Theorem 1.

Their analysis controls the stationarity proxy $\mathcal{G}_T = \frac{1}{T}\sum_t \mathbb{E}[\|\mathbf{g}_t\|_1/\sqrt{mn}]$, where $\mathbf{g}_t = \nabla f(\mathbf{X}_t)$, through the per-entry sign-error probabilities $q_{ij,t} = \Pr(\mathbf{S}_{t,ij} \neq \operatorname{sign}(\mathbf{g}_{t,ij}) \mid \mathbf{X}_t)$, in two steps. First, for an arbitrary sign oracle $\mathbf{S}_t$ they derive

$$\begin{aligned}\mathcal{G}_T \le\ & \frac{f(\mathbf{X}_0) - f^*}{\eta T} + \frac{L_*}{2}\,\eta \\ & + \underbrace{\frac{2}{T}\sum_t \mathbb{E}\left[\frac{1}{\sqrt{mn}}\sum_{i,j} |\mathbf{g}_{t,ij}|\, q_{ij,t}\right]}_{=:\,R_T}.\end{aligned} \tag{16}$$

Inequality (16) applies in particular to $\mathbf{S}_t = \operatorname{sign}(\operatorname{polar}(\mathbf{M}_t))$, but with the residual R_T left unbounded. Second, the rate is obtained by estimating R_T, and this estimate is derived *only* for the gradient-sign oracle $\mathbf{S}_t = \operatorname{sign}(\widetilde{\mathbf{G}}_t)$, where $\widetilde{\mathbf{G}}_t$ is an unbiased stochastic gradient with coordinatewise variance $\operatorname{Var}(\widetilde{\mathbf{G}}_{t,ij} \mid \mathbf{X}_t) \le \sigma^2_{ij}/n_b$ and n_b the mini-batch size. In that case a Markov–Jensen argument yields $|\mathbf{g}_{t,ij}|\, q_{ij,t} \le \sigma_{ij}/\sqrt{n_b}$, hence $R_T = \mathcal{O}(\|\sigma\|_1/\sqrt{mn\, n_b})$ with $\|\sigma\|_1 = \sum_{i,j}\sigma_{ij}$, which vanishes as $n_b \to \infty$.

This estimate bounds $q_{ij,t}$ by the noise level $\sigma_{ij}/\sqrt{n_b}$ and is therefore specific to $\operatorname{sign}(\widetilde{\mathbf{G}}_t)$, whose only source of sign error is gradient noise. It does not extend to the polar-sign oracle: even with exact gradients ($\sigma_{ij} = 0$) the latter satisfies $q_{ij,t} = \mathbf{1}[\operatorname{sign}(\operatorname{polar}(\mathbf{g}_t))_{ij} \neq \operatorname{sign}(\mathbf{g}_{t,ij})]$, which need not vanish. Consequently the finalized $\mathcal{O}(1/\sqrt{T})$ guarantee holds for the gradient-sign update $\mathbf{S}_t = \operatorname{sign}(\widetilde{\mathbf{G}}_t)$, whose direction never invokes the polar factor, and not for the LMO-based direction $\operatorname{sign}(\operatorname{polar}(\mathbf{M}_t))$ to which it is attributed.

On the instance of Theorem 1 the residual R_T is not merely unbounded: it exceeds $\mathcal{G}_T$. Consider the linear objective (11) with the 4×4 gradient $\mathbf{G}$ of Theorem 1. By Proposition 1 one has $\mathbf{g}_t \equiv \mathbf{G}$ and $\operatorname{polar}(\mathbf{M}_t) = \operatorname{polar}(\mathbf{G})$ at every iteration, so the oracle is deterministic and the inner-product identity $\mathbb{E}[\langle \mathbf{g}_t, \mathbf{D}_t\rangle \mid \mathbf{X}_t] = \frac{1}{\sqrt{mn}}\sum_{ij} |\mathbf{g}_{t,ij}|(1 - 2q_{ij,t})$ underlying (16) reduces to

$$\sum_{i,j} |\mathbf{G}_{ij}|\,(1 - 2q_{ij}) = \big\langle \mathbf{G}, \operatorname{sign}(\operatorname{polar}(\mathbf{G}))\big\rangle = -\tfrac{42468}{103} < 0. \tag{17}$$

Therefore $\sum_{ij} |\mathbf{G}_{ij}|\, q_{ij} > \frac{1}{2}\|\mathbf{G}\|_1$, and at every iteration

$$\frac{2}{\sqrt{mn}}\sum_{i,j} |\mathbf{G}_{ij}|\, q_{ij} \ > \ \frac{\|\mathbf{G}\|_1}{\sqrt{mn}} \ = \ \mathcal{G}_t. \tag{18}$$

The unestimated term R_T in (16) thus strictly exceeds the left-hand side, so the bound reduces to $\mathcal{G}_T \le (\text{nonnegative}) + R_T$ with $R_T > \mathcal{G}_T$ and imposes no constraint on $\mathcal{G}_T$. In agreement, (17) yields the exact per-step increase $f(\mathbf{X}_{t+1}) - f(\mathbf{X}_t) = -\eta\langle \mathbf{G}, \mathbf{D}_t\rangle = \frac{\eta}{4}\cdot\frac{42468}{103} > 0$ for every $\eta > 0$, which is the divergence established in Theorem 1.

A.7 Extended comparison with S-Muon

Taking the norms dual to the Ky Fan k-norms, Kravatskiy et al. (2025) obtain the *Fanion* family, whose updates $\sum_{i\le k}\mathbf{u}_i\mathbf{v}_i^\top$ interpolate between the rank-one step of the nuclear norm and Muon's full-rank $\mathbf{U}\mathbf{V}^\top$ at $k=\min(m,n)$; a conic combination of LMOs is again an LMO, for the norm dual to the corresponding combination of dual norms. Their S-Muon is one such combination, $\tau\,\mathbf{U}\mathbf{V}^\top+(1-\tau)\,c\,\mathrm{sign}(\mathbf{M}_t)$ with fixed $\tau\in[0,1]$, $c>0$ (their notation differs; we reserve α for compressor contraction and η for the learning rate). It is the sharpest available contrast with this paper: the sign enters *inside* the oracle, so the step is still an LMO for an explicit norm and inherits the convergence theory of one, where our three placements act *around* the oracle, on an object that is an LMO for no norm at all. Theorems 1–3 are the consequence. A caution from the same work applies to us directly: they exhibit an LMO method (rank-one Neon) markedly worse than Muon in practice despite sharing its convergence asymptotics in the bounds of Kovalev (2025) and Riabinin et al. (2025), from which our own guarantee descends. A rate of the form $\mathcal{O}(T^{-1/2})$ is a licence to run a method, not a prediction of the performance it will attain.

A.8 Proof of Theorem 4 (Divergence of EF21-SignMuon)

Recall from the main text that EF21-SignMuon (Algorithm A3) does not descend along the LMO output $\mathbf{D}_t=\mathrm{polar}(\tilde{\mathbf{M}}_t)$ itself, but along the error-feedback estimate $\mathbf{d}_t^{\mathrm{est}}$ of (13), followed by $\mathbf{X}_t=\mathbf{X}_{t-1}-\eta\,\mathbf{d}_t^{\mathrm{est}}$. A *single* magnitude α_t rescales the signs of *all* entries at once; this coupling is what the counterexample exploits. We now prove Theorem 4.

Proof idea Consider the target $\mathbf{D}_t$ as a 2×2 matrix whose off-diagonal is large and *flips sign every step*, while its diagonal is a small constant. The estimator (13) rescales all four signs by the same α_t, so the flipping off-diagonal holds α_t at $\Theta(1)$, which exceeds the small diagonal target. The diagonal of $\mathbf{d}_t^{\mathrm{est}}$ therefore never settles: it oscillates with $\Theta(1)$ amplitude, and its time-average has the sign *opposite* to the target. Being an average over an oscillation, this bias depends only on the oscillation's phase, which one-bit sign feedback cannot correct. It pushes one entry of $\mathbf{X}_t$ monotonically to infinity (Figure 1, right).

Two observations make the result independent of the parameters. First, the step size and smoothness only rescale the orbit, so we may fix $\eta=1$ and one value of L. Second, the divergence is due entirely to the estimator's response to a fixed target sequence; momentum affects only *how the gradients are sourced*, not the targets. A single explicit trajectory thus settles all $(L,\eta,\mu,\text{variant})$ at once. The mechanism is not degeneracy: the matrices driving the divergent tail have condition number $\frac{16}{9}$ throughout (the two-step preamble, which only sets the phase, is not claimed to be well conditioned).

Accordingly the proof has three parts. Part 1 rescales the problem to remove L and η. Part 2 isolates the dynamical core: for one fixed sequence of LMO targets, the estimator provably converges to a limit cycle whose diagonal has the wrong sign. Part 3 exhibits a concrete smooth function whose gradients generate that target sequence, for every μ and either momentum variant.

Proof

Part 1: rescaling.

Lemma 1 (Scale reduction) *If $\tilde f$ is $\tilde L$-smooth with EF21-SignMuon iterates $\tilde{\mathbf{X}}_t$ at step size 1, then $f(\mathbf{X}):=\frac{L\eta^2}{\tilde L}\,\tilde f(\mathbf{X}/\eta)$ is L-smooth and its run at step size η (same μ, same variant, from $\mathbf{0}$) satisfies $\mathbf{X}_t=\eta\,\tilde{\mathbf{X}}_t$ and $f(\mathbf{X}_t)=\frac{L\eta^2}{\tilde L}\,\tilde f(\tilde{\mathbf{X}}_t)$.*

Proof. $\nabla f(\mathbf{X})=\frac{L\eta}{\tilde L}\nabla\tilde f(\mathbf{X}/\eta)$, so f is L-smooth. Suppose $\mathbf{X}_s=\eta\tilde{\mathbf{X}}_s$ for $s<t$: the two gradients then differ by the positive factor $\frac{L\eta}{\tilde L}$, which $\mathbf{M}_t$ and $\tilde{\mathbf{M}}_t$ (positive combinations of past gradients) inherit and polar discards. Hence $\mathbf{D}_t$ and $\mathbf{d}_t^{\mathrm{est}}$ match the normalized run, and $\mathbf{X}_t=\eta\tilde{\mathbf{X}}_t$. ■ So it suffices to exhibit, for each $(\mu,\text{variant})$, a C^∞ $\tilde L$-smooth $\tilde f$ on which the *normalized* run ($\eta=1$) obeys (14); that $\tilde f$ may be taken bounded below (Assumption 1) is shown afterwards (Remark 1). We fix $\eta=1$ from now on.

Part 2: the estimator's limit cycle. The dynamical core of the proof is the behavior of the estimator (13) on the fixed target sequence

$$\mathbf{D}_1=\mathbf{S}_1,\quad \mathbf{D}_2=\mathbf{S}_2,\quad \mathbf{D}_t=\bar{\mathbf{D}}^{(-1)^t}\ (t\ge3), \tag{19}$$

$$\bar{\mathbf{D}}^{\pm}=\begin{pmatrix}7/25 & \pm24/25\\ \pm24/25 & -7/25\end{pmatrix},\qquad \mathbf{S}_1=\begin{pmatrix}-4/5 & 3/5\\ -3/5 & -4/5\end{pmatrix},\quad \mathbf{S}_2=\begin{pmatrix}3/5 & -4/5\\ 0 & 0\end{pmatrix}. \tag{20}$$

The divergence originates in the tail ($t\ge3$): the reflections $\bar{\mathbf{D}}^{\pm}$ share the diagonal $(\frac{7}{25},-\frac{7}{25})$, while their off-diagonal $\pm\frac{24}{25}$ reverses sign at every step. The rotation $\mathbf{S}_1$ and the rank-one $\mathbf{S}_2$ form a two-step preamble whose only role is to place the estimator in the divergent phase of the cycle.

Lemma 2 (Wrong-sign limit cycle) *On the targets* (19)*, the recursion* (13) *from $\mathbf{d}_0^{\mathrm{est}}=\mathbf{0}$ enters at $t=3$ the exact period-two cycle $\mathbf{d}_t^{\mathrm{est}}=\mathbf{d}_B$ (odd t), $\mathbf{d}_A$ (even t), where*

$$\mathbf{d}_B=\frac{1}{200}\begin{pmatrix}-61 & -201\\ -61 & -61\end{pmatrix},\qquad \mathbf{d}_A=\frac{1}{200}\begin{pmatrix}131 & -9\\ 131 & 131\end{pmatrix}. \tag{21}$$

Its $(2,2)$-entry averages $\frac12\big((\mathbf{d}_A)_{22}+(\mathbf{d}_B)_{22}\big)=+\frac{7}{40}$, opposite in sign to every target value $(\mathbf{D}_t)_{22}=-\frac{7}{25}$.

Proof. Substituting (19) into (13) gives the values

$$\begin{array}{c|cccc} t & 1 & 2 & 3 & 4 \text{ (then 2-periodic)} \\ \hline \alpha_t & \frac{7}{10} & \frac{21}{20} & \frac{131}{200} & \frac{24}{25} \\ \mathbf{d}_t^{\mathrm{est}} & \frac{1}{10}\left(\begin{smallmatrix} -7 & 7 \\ -7 & -7 \end{smallmatrix}\right) & \frac{1}{20}\left(\begin{smallmatrix} 7 & -7 \\ 7 & 7 \end{smallmatrix}\right) & \mathbf{d}_B & \mathbf{d}_A \end{array} \tag{22}$$

Each residual $\mathbf{D}_t - \mathbf{d}_{t-1}^{\mathrm{est}}$ has strictly nonzero entries, so the signs are unambiguous; e.g. at $t = 3$ it is $\frac{1}{100}\left(\begin{smallmatrix} -7 & -61 \\ -131 & -63 \end{smallmatrix}\right)$, all negative. From $t = 3$ the targets are 2-periodic and the pair $(\mathbf{d}_B, \mathbf{d}_A)$ reproduces itself: $\bar{\mathbf{D}}^+ - \mathbf{d}_B$ and $-(\bar{\mathbf{D}}^- - \mathbf{d}_A)$ are both entrywise positive with mean $\frac{24}{25}$, so $\mathbf{d}_B + \frac{24}{25}\mathbf{J} = \mathbf{d}_A$ and $\mathbf{d}_A - \frac{24}{25}\mathbf{J} = \mathbf{d}_B$ ($\mathbf{J}$ all-ones). ■

Since $\mathbf{X}_t = \mathbf{X}_{t-1} - \mathbf{d}_t^{\mathrm{est}}$ (recall $\eta = 1$), over one period the $(2, 2)$-coordinate changes by $-(\mathbf{d}_A + \mathbf{d}_B)_{22} = -\frac{7}{20}$. Consequently a term $-\gamma W_{22}$ in $\tilde{f}$, with $\gamma := \frac{7}{12}$, increases by $\gamma \cdot \frac{7}{20} = \frac{49}{240}$ per period. This is the divergence, provided a genuine smooth function produces the targets (19); Part 3 constructs one.

Part 3: realization by a smooth function. Fix (μ, variant) and set $\gamma = \frac{7}{12}$, $\nu = \frac{1}{1+2\mu}$, and

$$\begin{aligned} A &= \frac{1+\mu}{1-\mu} \text{ (standard)}, \\ A &= \frac{1+\mu}{(1-\mu)(1+2\mu)} \text{ (Nesterov)}. \end{aligned} \tag{23}$$

We build $\tilde{f} = g + \sum_{k=1}^{3} b_k$ from a periodic-plus-linear *field* g and three *localized corrections* b_k, all explicit.

The field is

$$g(\mathbf{W}) = -\gamma\, W_{22} + A\, \Phi_1(W_{12}) + A\, \Phi_2(W_{21}), \tag{24}$$

where $\Phi_i(w) = \int_0^w \psi_i$ and $\psi_i \colon \mathbb{R} \to [-1, 1]$ is a fixed C^∞, p_i-periodic function with $\int_0^{p_i} \psi_i = 0$,

$$p_1 = \tfrac{21}{20}, \quad p_2 = \tfrac{7}{20}, \qquad \delta = \tfrac{1}{100},$$

equal to $+1$ on $[\rho_i^+ - \delta, \rho_i^+ + \delta]$ and -1 on $[\rho_i^- - \delta, \rho_i^- + \delta]$ (mod p_i), where

$$\rho_1^+ = \tfrac{131}{200},\ \rho_1^- = \tfrac{140}{200}; \qquad \rho_2^+ = \tfrac{61}{200},\ \rho_2^- = 0.$$

The two bands are disjoint mod p_i (their centers are $\frac{9}{200} > 2\delta$ apart), so such a ψ_i exists; the zero-mean condition, met by balancing the rest of the period, makes Φ_i periodic (hence bounded). On each band $\Phi_i' = \psi_i = \pm 1$ exactly.

The corrections pin the first three gradients. Fix once a C^∞ cutoff $\phi \colon \mathbb{R} \to [0, 1]$ with $\phi \equiv 1$ on $[0, \frac{1}{2}]$ and $\phi \equiv 0$ on $[1, \infty)$, put $r = \frac{1}{50}$, and set

$$b_k(\mathbf{W}) = \langle \mathbf{C}_k,\, \mathbf{W} - \mathbf{Z}_k \rangle\, \phi\big(\|\mathbf{W} - \mathbf{Z}_k\|_F / r\big), \tag{25}$$

centered at the first three iterates (from Lemma 2)

$$\mathbf{Z}_1 = \mathbf{0}, \quad \mathbf{Z}_2 = \tfrac{1}{10}\left(\begin{smallmatrix} 7 & -7 \\ 7 & 7 \end{smallmatrix}\right), \quad \mathbf{Z}_3 = \tfrac{1}{20}\left(\begin{smallmatrix} 7 & -7 \\ 7 & 7 \end{smallmatrix}\right).$$

Each b_k is C^∞, supported in $\{\|\mathbf{W} - \mathbf{Z}_k\|_F \le r\}$; since its linear factor vanishes at $\mathbf{Z}_k$ while $\phi(0) = 1$, $\nabla b_k(\mathbf{Z}_k) = \mathbf{C}_k$. We choose $\mathbf{C}_k := \hat{\mathbf{G}}_k - \nabla g(\mathbf{Z}_k)$ with the explicit $\hat{\mathbf{G}}_k$ of (26) below, so that $\nabla \tilde{f}(\mathbf{Z}_k) = \hat{\mathbf{G}}_k$. Every term of $\tilde{f}$ has a bounded Hessian, so $\tilde{f}$ is C^∞ and $\tilde{L}$-smooth for a finite $\tilde{L}(\mu, \text{variant})$.

Lemma 3 (Realization) *For every $\mu \in [0, 1)$ and either variant, EF21-SignMuon on this $\tilde{f}$ ($\eta = 1$, from $\mathbf{0}$) generates exactly the targets* (19).

Proof. *The required gradients.* Momentum is a causal, positive-weight linear filter of the gradients, and we invert it. Define the *transient gradients*

$$\hat{\mathbf{G}}_t = \frac{\mathbf{M}_t - \mu \mathbf{M}_{t-1}}{1 - \mu} \quad (t = 1, 2, 3;\ \mathbf{M}_0 = \mathbf{0}) \tag{26}$$

and the *field gradient* $\mathbf{G}^\pm = \left(\begin{smallmatrix} 0 & \pm A \\ \pm A & -\gamma \end{smallmatrix}\right)$, where the prescribed buffer values $\mathbf{M}_{1,2,3}$ are given below. With $\bar{\mathbf{M}}^\pm = \left(\begin{smallmatrix} 0 & \pm 1 \\ \pm 1 & -\gamma \end{smallmatrix}\right)$, the factorization

$$\begin{gathered} \bar{\mathbf{M}}^\pm = \bar{\mathbf{D}}^\pm \left(\begin{smallmatrix} 24/25 & \mp 7/25 \\ \mp 7/25 & 337/300 \end{smallmatrix}\right) \\ (\text{second factor } \succ 0,\ \det = 1) \end{gathered} \tag{27}$$

shows $\mathrm{polar}(\bar{\mathbf{M}}^\pm) = \bar{\mathbf{D}}^\pm$, with singular values $\frac{3}{4}, \frac{4}{3}$ (the condition number $\frac{16}{9}$ of the idea).

Standard momentum ($\tilde{\mathbf{M}}_t = \mathbf{M}_t$). Take $\mathbf{M}_1 = \mathbf{S}_1, \mathbf{M}_2 = \mathbf{S}_2, \mathbf{M}_3 = \bar{\mathbf{M}}^-$; then $\hat{\mathbf{G}}_{1,2,3}$ are the explicit matrices (26). Feeding $\mathbf{G}^{(-1)^t}$ for $t \geq 4$ keeps $\mathbf{M}_t = \bar{\mathbf{M}}^{(-1)^t}$, because $(1-\mu)\mathbf{G}^{\pm} = \bar{\mathbf{M}}^{\pm} - \mu\bar{\mathbf{M}}^{\mp}$ with A as in (23). As $\operatorname{polar}(\mathbf{S}_1) = \mathbf{S}_1$ (orthogonal) and $\operatorname{polar}(\mathbf{S}_2) = \mathbf{S}_2$ (rank one), (27) yields the targets (19).

Nesterov momentum ($\tilde{\mathbf{M}}_t = (1+\mu)\mathbf{M}_t - \mu\mathbf{M}_{t-1}$). Take $\mathbf{M}_1 = \frac{1}{1+\mu}\mathbf{S}_1\mathbf{H}_1$, $\mathbf{M}_2 = \frac{1}{1+\mu}(\mathbf{S}_2 + \mu\mathbf{M}_1)$, $\mathbf{M}_3 = \bar{\mathbf{M}}'^-$ and $\mathbf{M}_t = \bar{\mathbf{M}}'^{(-1)^t}$ ($t \geq 4$), where $\bar{\mathbf{M}}'^{\pm} = \left(\begin{smallmatrix} 0 & \pm\nu \\ \pm\nu & -\gamma \end{smallmatrix}\right)$, $\mathbf{H}_1 = \operatorname{diag}(1, 1+T)$, and, for $\mu > 0$,

$$T = \left(\tfrac{140(1+\mu)^2}{1+2\mu} - 44\right)\tfrac{1+\mu}{117\mu} > 0.$$

(At $\mu = 0$ the Nesterov rule reads $\tilde{\mathbf{M}}_t = \mathbf{M}_t$ and is the standard case already treated, so nothing is left to prove there.) Then $\tilde{\mathbf{M}}_1 = \mathbf{S}_1\mathbf{H}_1$ and $\tilde{\mathbf{M}}_2 = \mathbf{S}_2$ have polar factors $\mathbf{S}_1, \mathbf{S}_2$, and $\tilde{\mathbf{M}}_t = \bar{\mathbf{M}}^{(-1)^t}$ for $t \geq 4$ (since $\nu(1+2\mu) = 1$). The one nontrivial step is $t = 3$: with $\mathbf{H}_3 := \bar{\mathbf{D}}^-\tilde{\mathbf{M}}_3$ (so $\tilde{\mathbf{M}}_3 = \bar{\mathbf{D}}^-\mathbf{H}_3$, the reflection being an involution), the stated T is exactly the value making $\mathbf{H}_3$ symmetric, and $\mathbf{H}_3 \succ 0$ for all $\mu \in (0,1)$: under $\mu = \frac{s}{1+s}$ ($s > 0$) the numerators of its two leading minors are polynomials in s with nonnegative coefficients. Hence $\operatorname{polar}(\tilde{\mathbf{M}}_3) = \bar{\mathbf{D}}^-$, completing (19).

The function delivers these gradients. It remains to verify $\nabla\tilde{f}(\tilde{\mathbf{X}}_{t-1}) = \hat{\mathbf{G}}_t$ for $t \leq 3$ and $= \mathbf{G}^{(-1)^t}$ for $t \geq 4$. By Lemma 2 the iterates $\tilde{\mathbf{X}}_0, \tilde{\mathbf{X}}_1, \tilde{\mathbf{X}}_2$ are exactly the centers $\mathbf{Z}_1, \mathbf{Z}_2, \mathbf{Z}_3$, where $\nabla\tilde{f} = \hat{\mathbf{G}}_{1,2,3}$ by construction. The three balls are disjoint ($\|\mathbf{Z}_j - \mathbf{Z}_k\|_F \geq \frac{7}{10} > 2r$), and every later iterate has $(1,2)$-entry $\geq \frac{131}{200}$ while the centers have it ≤ 0, so no ball is re-entered. For $t \geq 4$ the query lies in the field, where

$$\nabla g(\mathbf{W}) = \begin{pmatrix} 0 & A\psi_1(W_{12}) \\ A\psi_2(W_{21}) & -\gamma \end{pmatrix}.$$

The period shift $\tilde{\mathbf{X}}_{t+2} - \tilde{\mathbf{X}}_t = -(\mathbf{d}_A + \mathbf{d}_B)$ advances W_{12} by $+p_1$ and W_{21} by $-p_2$ (one full period each), so the off-diagonal coordinates keep the residues ρ_i^+ at odd indices and ρ_i^- at even indices; there $\psi_i = \pm 1$, giving $\nabla g = \mathbf{G}^{(-1)^t}$. ■

Proof of Theorem 4. By Lemma 3 the normalized run produces the targets (19), so by Lemma 2 its estimator locks onto the wrong-sign cycle and $\tilde{\mathbf{X}}_{t+2} - \tilde{\mathbf{X}}_t$ is a constant shift. Along it Φ_1, Φ_2 return to their values and the corrections b_k vanish, so only the linear term acts: $\tilde{f}(\tilde{\mathbf{X}}_{t+2}) - \tilde{f}(\tilde{\mathbf{X}}_t) = -\gamma(-\frac{7}{20}) = \frac{49}{240} > 0$. By Lemma 1, f then obeys (14) with $c = \frac{49}{240\tilde{L}}$. As $(L, \eta, \mu, \text{variant})$ were arbitrary, no rule $\eta = \eta(L, \mu)$ can prevent divergence. ■

Remark 1 (Boundedness below) *Only the linear term $-\gamma W_{22}$ makes $\tilde{f}$ unbounded below, and only as $W_{22} \to +\infty$, a region the iterates never enter, since $(\tilde{\mathbf{X}}_t)_{22} \leq \frac{7}{10}$ throughout (it decreases after the transient). Replacing $-\gamma W_{22}$ by any C^∞ function that agrees with it on $\{W_{22} \leq 1\}$ and is constant on $\{W_{22} \geq 2\}$ therefore leaves the whole trajectory — and* (14) *— unchanged while making $\tilde{f}$ bounded below (Assumption 1); the theorem is stated with this modification in force.*

Remark 2 (Verification) *The construction is checked in two independent ways: symbolically, in exact rational arithmetic, and numerically, by running the float64 reference implementation of Algorithm A3 on the assembled $\tilde{f}$. The right panel of Figure 1 confirms that at $\mu = 0$ EF21-SignMuon is the only one of the eight methods that diverges, the others (SignMuon, MuonUSign, MuonSign, EF21-MuonUSign, EF21-MuonSign, SignSGD, Muon) staying bounded; Figure A1 confirms that EF21-SignMuon diverges at the exact rate $\frac{49}{480}$ for every $\mu \in \{0, \frac{1}{2}, 0.9, 0.95, 0.99\}$ under both standard and Nesterov momentum, as the reduction predicts (`code/counterexamples/problems.py`, `code/counterexamples/plot_ef21_momentum.py`).*

A.9 Convergence of EF21-MuonUSign and EF21-MuonSign

We do not analyse the error-feedback methods from scratch. EF21-MuonUSign and EF21-MuonSign are *exact instances* of EF21-Muon (Gruntkowska et al. 2025, Algorithm 3), already analysed in the layer-wise, stochastic, federated setting. Three things must be checked before its guarantees transfer: our step is their LMO step, our loop is their loop (Proposition 2), and our messages come from contractive compressors (Lemma 4). Only the last is non-trivial. Table A1 is the change of variables.

Notation and constants. For the layer tuple $\mathbf{X} = [\mathbf{X}_1, \ldots, \mathbf{X}_p]$, $\mathbf{X}_i \in \mathbb{R}^{m_i \times n_i}$, of the Problem Statement write $d_i := m_i n_i$, $r_i := \min(m_i, n_i)$, $d_{\max} := \max_i d_i$; a second subscript selects a layer ($\mathbf{X}_{t,i}, \mathbf{G}_{t,i}$). We use $\|\mathbf{Y}\|_{2\to 2} \leq \|\mathbf{Y}\|_F \leq \|\mathbf{Y}\|_* \leq \sqrt{\operatorname{rank}\mathbf{Y}}\,\|\mathbf{Y}\|_F$. Smoothness constants: L_i for f and $L_{i,j}$ for f_j in Assumption 2, with $\tilde{L}_i^2 := \frac{1}{N}\sum_j L_{i,j}^2$ and $L := \max_i L_i$; in the (L^0, L^1) form the pairs are again per layer for f and per layer and client for f_j, with $L^1_{i,\max} := \max_j L^1_{i,j}$.

Assumption 1 (Stochastic gradient) *Each client's stochastic gradient is unbiased, $\mathbb{E}_\xi[\nabla f_j(\mathbf{X};\xi)] = \nabla f_j(\mathbf{X})$, with bounded variance $\mathbb{E}_\xi\|\nabla f_j(\mathbf{X};\xi) - \nabla f_j(\mathbf{X})\|_*^2 \leq \sigma^2$.*

Assumptions 1–1 are Assumptions 1–2 and 6–10 of Gruntkowska et al. (2025) with the layer norms taken spectral; our variance bound is stated in the nuclear norm and implies theirs via $\|\cdot\|_F \leq \|\cdot\|_*$. The clause $f_j \geq f_j^*$ of Assumption 1 is needed only for Corollary 2.

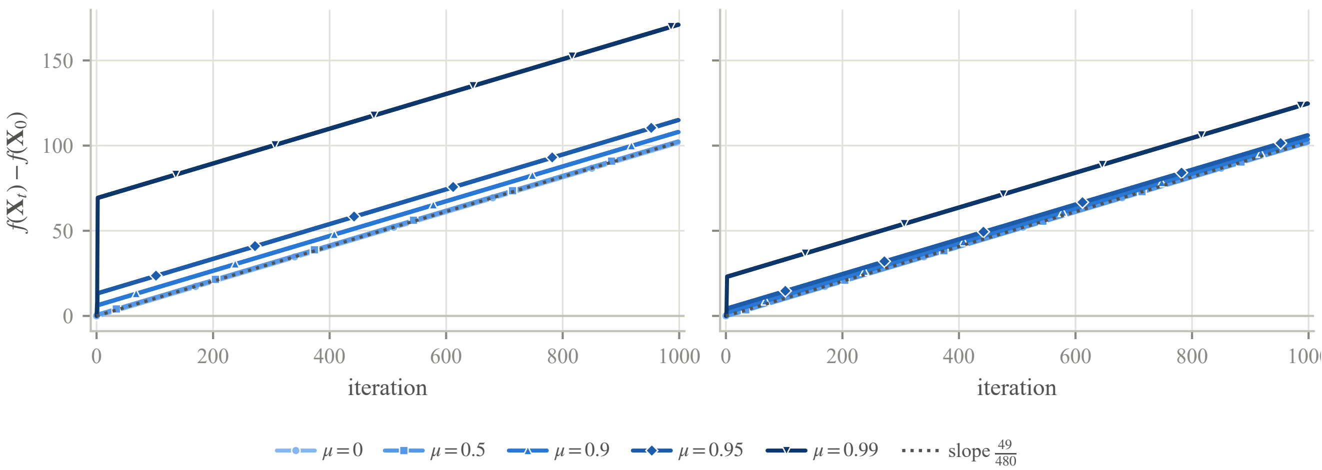


Figure A1: Momentum does not prevent the divergence of EF21-SignMuon. For each momentum coefficient μ and variant, EF21-SignMuon is run on the corresponding instance of Theorem 4 and $f(\mathbf{X}_t) - f(\mathbf{X}_0)$ is plotted; *left:* standard momentum, *right:* Nesterov. Every setting diverges at the common rate $\frac{49}{480}$ (dotted). Subtracting $f(\mathbf{X}_0)$ removes the only genuinely μ-dependent offset (the field constant $A(\mu)$ scales a bounded periodic term); what remains is a bounded transient, largest as $\mu \to 1$, on top of the shared linear divergence.

Main result

Theorem 5 (Convergence of the EF21 methods) *Run the federated Algorithm A9 with the EF21 uplink and EMA momentum $\mu \in [0, 1)$. Then:*

(i) (smooth; both methods) *under Assumptions 1–1, with the sharp learning rate $\eta_{t,i} = \gamma_i \|\mathbf{G}_{t,i}\|_*$ and tuned (γ_i, μ), both reach $\frac{1}{T}\sum_{t<T} \mathbb{E}\|\nabla f(\mathbf{X}_t)\|_*^2 = \mathcal{O}(T^{-1/2})$ (Corollary 1);*

(ii) (generalized smooth; uplink only) *under (L^0, L^1)-smoothness, EF21-MuonUSign with a plain* constant *learning rate reaches $\min_{t \le T} \sum_i \mathbb{E}\|\nabla_i f(\mathbf{X}_t)\|_* = \mathcal{O}(T^{-1/4})$ (Corollary 2).*

Both statements are written for a per-layer constant common to all layers ($\gamma_i \equiv \gamma$, $\eta_i \equiv \eta$); for unequal constants the left-hand sides carry the step-size weights of Corollaries 1–2. The majority-vote placements (Theorems 1–3) and EF21-SignMuon (Theorem 4) admit no such guarantee.

Part (i) is the rate announced in the main text (via $\min_t \le$ average); the centralized Algorithms A6–A7, and hence $\mathbf{X}_t$ in (i), are the case $N = 1$.

The reduction

The step is their LMO step. For $\mathbf{G} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ the framework's oracle over the spectral ball of radius τ is $\mathrm{lmo}(\mathbf{G}) = \mathbf{X} - \tau\mathbf{U}\mathbf{V}^\top$, which is our server step with $\tau = \eta_{t,i}$, since $\mathbf{D}_{t,i} = -A(\mathbf{G}_{t,i}) = \mathbf{U}_{t,i}\mathbf{V}_{t,i}^\top$:

$$\mathbf{X}_{t,i} = \mathbf{X}_{t-1,i} - \eta_{t,i}\,\mathbf{D}_{t,i} = \mathrm{lmo}_{\mathcal{B}(\mathbf{X}_{t-1,i}, \eta_{t,i})}(\mathbf{G}_{t,i}). \tag{28}$$

Their "sharp" step $\mathbf{X} - \gamma\,\mathbf{G}^\sharp$ uses $\mathbf{G}^\sharp = \|\mathbf{G}\|_* \mathbf{U}\mathbf{V}^\top$, so a constant γ_i *is* the schedule $\eta_{t,i} = \gamma_i \|\mathbf{G}_{t,i}\|_*$ and a constant $\eta_{t,i}$ is the plain Muon rate: both are learning-rate choices, not algorithmic changes. The spectral norm-equivalence constants are $\underline{\rho}_i = 1$, $\bar{\rho}_i = \sqrt{r_i}$. At rank-deficient $\mathbf{G}$ the LMO is non-unique, but any selection works: the analysis uses only $\langle \mathbf{G}, \mathrm{lmo}(\mathbf{G})\rangle = -\tau\|\mathbf{G}\|_*$ and $\|\mathrm{lmo}(\mathbf{G})\|_{2\to 2} \le \tau$, with $\mathbf{G} = \mathbf{0}$ read as the zero step ($\mathbf{0}^\sharp = \mathbf{0}$).

The momentum is their momentum. Algorithm A9 uses $\mathbf{M}_t = \mu\mathbf{M}_{t-1} + (1-\mu)\mathbf{G}_t$, the framework's momentum with $\beta = 1 - \mu$. From zero initialization the heavy-ball form of (7) differs only by the constant factor $1/(1-\mu)$ on the momentum stream; the EF21 recursion is positively homogeneous and the Muon LMO scale-invariant, so that factor leaves the iterates unchanged under a constant $\eta_{t,i}$ and is absorbed into γ_i under the sharp schedule. (The Nesterov branch is a different filter; see the end of this appendix.)

Proposition 2 (Exact instance) *Fix μ, a learning-rate schedule, and the LMO selection above. Started from $\mathbf{M}_0^{(j)} = \mathbf{g}_0^{(j)} = \mathbf{G}_0 = \mathbf{0}$, $\mathbf{W}_0 = \mathbf{X}_0$, Algorithm A9 with the EF21 uplink and either downlink mode ($\mathcal{C}^\downarrow \in \{$exact, EF21-P$\}$) produces the same trajectory as Algorithm 3 of Gruntkowska et al. (2025) with spectral norms, scaled-sign worker compressors, identity/scaled-sign server compressor, $\beta = 1 - \mu$, and radii $t_i^k = \eta_{k,i}$ – up to a one-step index shift $X^{t+1} = \mathbf{X}_t$.*

	This paper	**EF21-Muon**
clients / rounds	N, T	$n, K = T+1$
iterate	$\mathbf{X}_t$	X^{t+1}
momentum	μ	$1-\beta$
learning rate	$\eta_{t,i}$	radius t_i^t
norm equivalence	$1,\ \sqrt{r_i}$	$\underline{\rho}_i,\ \bar{\rho}_i$
uplink compr.	scaled sign	$\mathbb{B}_2(\alpha_D)$
downlink compr.	exact / sign	$\mathcal{I}$ / $\mathbb{B}_2(\alpha_P)^\dagger$
compression α	$1/d_{\max}$	α_D

Table A1: Change of variables from our notation to that of Gruntkowska et al. (2025). Layer norms $\|\cdot\|_{(i)}, \|\cdot\|_{(i)\star}$ are read as $\|\cdot\|_{2\to 2}, \|\cdot\|_*$ and smoothness constants $L_i^0, \tilde{L}_i^0$ as $L_i, \tilde{L}_i$; the index shift is Proposition 2. †Their Theorem 19 as printed asks for a server compressor in $\mathbb{B}(\alpha_P)$, contractive in the *layer* norm; the Euclidean class $\mathbb{B}_2(\alpha_P)$ that the scaled sign belongs to is licensed only by their Remark 23, at the $\bar{\rho}_i^2$ price charged in Corollary 1 and Remark 4.

Proof The loops differ only in where the round is cut: they order it *step* → *downlink* → *gradient* → *uplink*, we order it *downlink* → *gradient* → *uplink* → *step*. Their iteration $k = 0$ is vacuous under our initialization: $\mathbf{G}_0 = \mathbf{0}$ gives $X^1 = X^0 = \mathbf{X}_0$ and a zero downlink residual, so $W^1 = \mathbf{X}_0$. Thereafter their iteration $k = t$ performs our round t verbatim – same momentum, same compressed residual, same LMO step (28) – giving $X^{t+1} = \mathbf{X}_t$, $W^{t+1} = \mathbf{W}_t$ by induction. Running their method for $K = T+1$ iterations therefore yields $\{\mathbf{X}_0, \mathbf{X}_0, \mathbf{X}_1, \ldots, \mathbf{X}_{T-1}\}$, and any average or minimum over these equals ours up to one duplicated nonnegative term. ■

Zero initialization also makes their initial-error constant Ψ^0 explicit: the $\|\mathbf{M}_0 - \mathbf{g}_0\|$ term vanishes and the gradient-deviation terms become $\|\nabla_i f(\mathbf{X}_0)\|$. The β^{-1} surviving in Ψ^0 enters only through the Ψ^0/T term, which $T^{-1/2}$ dominates.

The one real check: the scaled sign is a compressor The framework needs every transmitted message to come from a *contractive compressor*: a map with $\mathbb{E}\|\mathcal{C}(\mathbf{Y}) - \mathbf{Y}\|^2 \le (1-\alpha)\|\mathbf{Y}\|^2$ for some $\alpha \in (0, 1]$ (Gruntkowska et al. 2025, Def. 1). This is exactly what fails for a bare sign and holds once it is scaled.

Lemma 4 (Contractivity of the scaled sign) *For every $\mathbf{Y} \in \mathbb{R}^{m\times n}$ ($d = mn$), the scaled sign $\mathcal{C}(\mathbf{Y}) = \operatorname{mean}(|\mathbf{Y}|)\operatorname{sign}(\mathbf{Y})$, with exact zeros resolved to ± 1 as in Section 4, satisfies*

$$\|\mathcal{C}(\mathbf{Y}) - \mathbf{Y}\|_F^2 = \|\mathbf{Y}\|_F^2 - \tfrac{1}{d}\|\mathbf{Y}\|_1^2 \le \big(1 - \tfrac{1}{d}\big)\|\mathbf{Y}\|_F^2, \tag{29}$$

so $\mathcal{C}$ is Euclidean-contractive with $\alpha = 1/d$. The identity holds for every draw of the random signs, not merely in expectation, and gives the exact contraction $\alpha(\mathbf{Y}) = \|\mathbf{Y}\|_1^2/(d\|\mathbf{Y}\|_F^2)$, which is $\Theta(1)$ for dense $\mathbf{Y}$ (e.g. $\to 2/\pi$ for i.i.d. Gaussian entries) and equals $1/d$ exactly at any 1-sparse $\mathbf{Y}$.

Proof Write $c := \|\mathbf{Y}\|_1/d$ and $s_{kl} = \pm 1$ for the transmitted signs. A nonzero entry contributes $(|Y_{kl}| - c)^2$ and a zero entry contributes $(c\, s_{kl})^2 = c^2$ whichever sign was drawn, so $\|\mathcal{C}(\mathbf{Y}) - \mathbf{Y}\|_F^2 = \|\mathbf{Y}\|_F^2 - 2c\|\mathbf{Y}\|_1 + c^2\|\mathbf{Y}\|_0 + c^2(d - \|\mathbf{Y}\|_0) = \|\mathbf{Y}\|_F^2 - 2c\|\mathbf{Y}\|_1 + dc^2$, which is (29) on substituting c. The bound then follows from $\|\mathbf{Y}\|_1 \ge \|\mathbf{Y}\|_F$, with equality exactly at the 1-sparse $\mathbf{Y}$; the Gaussian limit uses $\mathbb{E}|Y_{kl}| = \sqrt{2/\pi}\,\sigma$. ■

The $\|\mathbf{Y}\|_0$ terms cancelling is the convention paying for itself: under $\operatorname{sign}(0) = 0$ the same computation leaves a sparsity-dependent identity, and $1/d$ becomes an unattained infimum.

The scaling is what carries the lemma. A bare sign contracts for no α at all: as $\|\mathbf{Y}\|_F \to 0$ on a fixed support, $\|\operatorname{sign}(\mathbf{Y}) - \mathbf{Y}\|_F \to \sqrt{\|\mathbf{Y}\|_0}$. That is the line between our divergent and convergent methods: the majority-vote methods transmit unscaled signs of full quantities, the EF21 variants the scaled sign of a *residual*, at the cost of one extra scalar per layer per round. Per layer the uplink scaled sign lies in $\mathbb{B}_2(1/d_i) \subseteq \mathbb{B}_2(\alpha)$ with $\alpha := 1/d_{\max}$; the downlink is exact for EF21-MuonUSign ($\alpha_P = 1$) and the same scaled sign for EF21-MuonSign.

Transferred guarantees All requirements hold, so the framework's theorems apply through Table A1. We state the rates and stepsize rules; the explicit non-asymptotic bounds are Gruntkowska et al. (2025), Theorem 19 (smooth) and Theorem 24 ((L^0, L^1)), at these constants.

Corollary 1 (Smooth case; both methods) *Let Assumptions 1–1 hold. Run Algorithm A9 with the EF21 uplink, EMA momentum, and $\eta_{t,i} = \gamma_i\|\mathbf{G}_{t,i}\|_*$ with any γ_i below the per-layer threshold of Gruntkowska et al. (2025, Thm. 19) under Table A1. Then, applying the momentum tuning of Gruntkowska et al. (2025, Cor. 2) per layer (their Corollaries 1–2 state it for the layer-wise initialization and for $p = 1$ respectively; the combination is routine),*

$$\tfrac{1}{T}\textstyle\sum_{t<T}\sum_{i=1}^{p} w_i\, \mathbb{E}\|\nabla_i f(\mathbf{X}_t)\|_*^2 = \mathcal{O}(T^{-1/2}),$$

with $w_i := \gamma_i/(\frac{1}{p}\sum_l \gamma_l)$; for a common γ_i all $w_i = 1$ and the left side is $\frac{1}{T}\sum_{t<T}\mathbb{E}\|\nabla f(\mathbf{X}_t)\|_*^2$.

Proof By Proposition 2 the run is an instance of Algorithm 3 with $K = T + 1$, and by Lemma 4 its compressors satisfy $\alpha_D = \alpha$ and $\alpha_P \in \{1, \alpha\}$ (EF21-MuonSign's Euclidean downlink is admissible by their Remark 23, which multiplies the α_P-terms of the threshold constant ζ_i by $\bar{\rho}_i^2 = r_i$). Substituting $\underline{\rho}_i = 1, \bar{\rho}_i = \sqrt{r_i}$ into Gruntkowska et al. (2025, Thm. 19) gives the threshold and the rate; the duplicated $\mathbf{X}_0$-term costs a factor $\frac{T+1}{T}$. ■

Since $\gamma_i \lesssim \zeta_i^{-1/2}$, the threshold shrinks polynomially in the layer dimension – through $\bar{\rho}_i = \sqrt{r_i}$ and the uplink $\alpha = 1/d_{\max}$ – and for EF21-MuonSign by a further $\sqrt{r_i}$ from the compressed downlink. That extra factor is structural: the scaled sign is spectrally contractive for no parameter at all (Remark 4).

Corollary 2 **((L^0, L^1) case; EF21-MuonUSign)** *Let Assumption 2 hold in its (L^0, L^1) form. Run EF21-MuonUSign (exact downlink) with $S := T + 2$, momentum $\mu = 1 - S^{-1/2}$, and the* constant *per-layer rate $\eta_{t,i} \equiv \eta_i / S^{3/4}$ for any $\eta_i \leq 1$ with $\eta_i^2 \sqrt{r_i}\, (L^1_{i,\max})^2 = \mathcal{O}(1)$.*[1] *Then*

$$\min_{0 \leq t \leq T} \sum_{i=1}^p v_i\, \mathbb{E}\|\nabla_i f(\mathbf{X}_t)\|_* = \mathcal{O}(T^{-1/4}), \quad v_i := \frac{\eta_i}{\frac{1}{p}\sum_l \eta_l}.$$

Proof Their Theorem 24 requires the identity server compressor, i.e. the exact downlink of EF21-MuonUSign (so EF21-MuonSign is excluded). Applying it with $K = T + 1$, $\beta = S^{-1/2}$ and Table A1 gives the schedule and the rate. ■

Under plain smoothness ($L^1 = 0$) every constraint above collapses to $\eta_i \leq 1$. The *constant*-learning-rate EF21-MuonUSign we actually run therefore already converges at $\mathcal{O}(T^{-1/4})$; the sharp schedule yields only the faster $\mathcal{O}(T^{-1/2})$ in squared norm.

Scope The reduction covers the two error-feedback methods and no others, which matches our negative results. The majority-vote methods send unscaled signs and fall outside the framework. EF21-SignMuon compresses the LMO *output* rather than the gradient, so it tracks the non-Lipschitz polar factor and the momentum-tracking step breaks (Theorem 4). EF21-MuonSign gets the smooth guarantee but not the (L^0, L^1) one, whose theory assumes an uncompressed downlink.

The remaining gap is the Nesterov branch, which our language-model runs use (Appendix A.16): with $\beta := 1 - \mu$ it steers by $\tilde{\mathbf{M}}_t = (1 - \beta)\mathbf{M}_t + \beta \mathbf{G}_t$ rather than by the buffer $\mathbf{M}_t$, a two-tap filter of the gradients where the framework wants a first-order recursion. The discrepancy is small and explicit: writing $\mathbf{n}_t := \mathbf{G}_t - \nabla_i f(\mathbf{X}_t)$, we have $\nabla_i f - \tilde{\mathbf{M}}_t = (1-\beta)(\nabla_i f - \mathbf{M}_t) - \beta\, \mathbf{n}_t$, and since $\mathbb{E}\langle \nabla_i f - \mathbf{M}_t, \mathbf{n}_t \rangle = -\beta\, \mathbb{E}\|\mathbf{n}_t\|_2^2$,

$$\mathbb{E}\|\nabla_i f - \tilde{\mathbf{M}}_t\|_2^2 \leq (1 - \beta)^2\, \mathbb{E}\|\nabla_i f - \mathbf{M}_t\|_2^2 + 3\beta^2 \sigma_i^2.$$

The first term is exactly the deviation Gruntkowska et al. (2025, Thm. 19) already tracks, contracted rather than enlarged; the second is dominated by the $\beta\sigma_i^2$ term already in its bound. Nesterov should therefore degrade the constants rather than the rate. We state this as an expectation, not as a corollary, since the tracking recursion for $\|\tilde{\mathbf{M}}_{t+1} - \tilde{\mathbf{M}}_t\|$ would have to be redone as well.

Remark 3 (The cost of one bit) *The uplink $\alpha = 1/d_{\max}$ is what makes the threshold of Corollary 1 shrink with the dimension – the worst-case cost of one-bit messages, as for Top-*1 *compressors in Euclidean EF21 (Richtárik, Sokolov, and Fatkhullin 2021).* On the uplink *it is worst-case only: it needs an all-but-one-coordinate-negligible residual, whereas the momentum residual is dense with $\alpha(\mathbf{\Delta}) = \Theta(1)$ (Lemma 4), and the framework is stated to extend straightforwardly to iteration-dependent α (Gruntkowska et al. 2025, Rem. 12).*

The downlink of EF21-MuonSign does not inherit this. *Its residual is not a gradient difference refreshed by fresh stochasticity, but the output of the compressor's own recursion*

$$\mathbf{\Delta}_{t+1}^{\downarrow} = \mathbf{\Delta}_t^{\downarrow} - \eta_{t+1}\mathbf{D}_{t+1} - \operatorname{mean}|\mathbf{\Delta}_t^{\downarrow}|\operatorname{sign}(\mathbf{\Delta}_t^{\downarrow}),$$

in which every coordinate is corrected by the same *amount. Coordinates driven harder than that mean are never caught while the bulk holds the mean down, so the residual concentrates – and concentration is what drives α down. We observe it: $\alpha(\mathbf{\Delta}^{\downarrow})$ falls to 1.2×10^{-4} on the layers built from a zero initialization, nearly four orders of magnitude below its own uplink on the same layers (Section 5.3). That is still far above the $1/d = 4.2 \times 10^{-7}$ floor, so the dimension dependence of Corollary 1 is not attained; what is lost is the $\Theta(1)$ contraction that makes it harmless upstream.*

Remark 4 (Why the sharp route is closed) *Corollary 1 reaches EF21-MuonSign only through the $\bar{\rho}_i^2$-penalized branch of Gruntkowska et al. (2025, Rem. 23), and this is forced. The sharp branch asks for a server compressor in $\mathbb{B}(\alpha_P)$, contractive in the* layer *norm, and the scaled sign is in no such class for any $\alpha_P > 0$. Take $\mathbf{Y}_\delta = (1-\delta)\mathbf{I}_n + \delta\mathbf{J}_n$, with $\mathbf{J}_n$ all ones, $0 < \delta < 1$ and $n \geq 2$. Every entry is positive, so* $\operatorname{sign}(\mathbf{Y}_\delta) = \mathbf{J}_n$ *and* $\operatorname{mean}|\mathbf{Y}_\delta| = (1 + (n-1)\delta)/n$, *whence*

$$\mathcal{C}(\mathbf{Y}_\delta) - \mathbf{Y}_\delta = (1-\delta)\big(\tfrac{1}{n}\mathbf{J}_n - \mathbf{I}_n\big),$$

with eigenvalues 0 *and* $-(1-\delta)$. *So $\|\mathcal{C}(\mathbf{Y}_\delta) - \mathbf{Y}_\delta\|_{2\to2} = 1 - \delta$ against $\|\mathbf{Y}_\delta\|_{2\to2} = 1 + (n-1)\delta$, a ratio tending to 1 as $\delta \downarrow 0$: no $\alpha_P > 0$ survives, on any layer of rank above one. The framework's sufficient condition is unattainable for the same*

[1] These are the binding conditions of Gruntkowska et al. (2025, Thm. 24); the two remaining ones relax as T grows. Their printed second condition is tighter than the display its own proof establishes by a factor $K + 1$; we quote the latter.

Experiment	GPU	CPU	RAM (GB)	OS	Python	PyTorch / CUDA
to be filled: one row per (experiment, machine) from `python3 -m common.hardware --scan results --latex`						

Table A2: Computing infrastructure, per experiment. Generated from the per-run hardware records rather than written by hand; where one experiment was run on more than one machine, it contributes one row per machine.

reason – by the norm-equivalence argument of Gruntkowska et al. (2025, App. D) a compressor in $\mathbb{B}_2(\alpha)$ is certified to lie in $\mathbb{B}(\alpha_P)$ only if $\alpha > 1 - 1/r_i$, i.e. $\alpha > 0.9987$ for our 768×3072 matrices, against an isotropic $2/\pi$. The factor $\bar{\rho}_i^2 = r_i$ in ζ_i is therefore not removable by a sharper analysis of this compressor, only by changing it – to random dropout ($\alpha_P = p$) or a Top-K SVD compressor ($\alpha_P = 1 - \sigma_{K+1}^2/\sigma_1^2$), both layer-norm contractive at a comparable budget (Gruntkowska et al. 2025, App. D), and neither strictly one-bit.

Remark 5 (From Muon to Gluon) *Only $(\underline{\rho}_i, \bar{\rho}_i) = (1, \sqrt{r_i})$ is spectral-norm-specific: Lemma 4 is Euclidean and the framework's theorems hold for arbitrary layer norms. A new geometry has to supply only its LMO and its norm-equivalence pair; both corollaries then hold with those constants in place of $(1, \sqrt{r_i})$, giving* EF21-GluonUSign *and* EF21-GluonSign *for the Gluon setting (Riabinin et al. 2025). Admissible geometries abound – $\ell_1 \to \ell_\infty$ for embeddings (Pethick et al. 2025), the Schatten-p norms (Cesista 2025), the Ky Fan duals of the Fanion family (Kravatskiy et al. 2025) – and, by the closure property of the last work, conic combinations of them, with $\bar{\rho}_i \le \left(\sum_j \alpha_j / \bar{\rho}_i^{(j)}\right)^{-1}$ for the mixture $\|\mathbf{X}_i\| = \sum_j \alpha_j \|\mathbf{X}_i\|_{(j)}$. Convergence is thus a property of error feedback plus scaled-sign compression, not of the spectral geometry. The two ways to bring a sign into a Muon step stay cleanly apart: inside the oracle it changes the norm and the guarantee follows the norm; outside it, it is a compressor and the guarantee has to come from error feedback.*

We assume the exact spectral LMO, as do all analyses of Muon-type methods (Li and Hong 2025; Kovalev 2025; Riabinin et al. 2025; Gruntkowska et al. 2025); in practice it is the five-step iteration of Algorithm A1 (Amsel et al. 2026; Grishina, Smirnov, and Rakhuba 2025), with vector parameters and the last layer trained by AdamW as usual (Jordan et al. 2024). That substitution is not covered by the reduction either, but it is the mildest of the gaps: Shulgin et al. (2025) analyse the inexact Muon update directly and find the method tolerant of oracle error, which is consistent with our runs: the five-step approximation reproduces record #40 to within seed noise (Table 3).

A.10 Reproducibility Details

Computing infrastructure. The experiments in this paper were not all run on the same machine, so Table A2 reports the hardware per experiment rather than as a single sentence. Each row is generated from the record every run stamps into its own output at the moment it runs (`python3 -m common.hardware --scan results --latex`), so the table describes the machines the reported numbers actually came from and not the machine that happened to typeset the paper. Exact package versions are pinned in `code/requirements.txt`.

Randomness and seeds. Seeding goes through a `seed_everything` routine that seeds the Python `random` module, NumPy, and PyTorch (including all CUDA devices) and enables deterministic cuDNN kernels (`cudnn.deterministic=True`, `cudnn.benchmark=False`). The federated experiment of Table 2 reports five seeds (0–4) per method and the centralized experiment of Table 1 three (0–2), each quoted as mean $\pm$ one sample standard deviation across seeds; the nanoGPT runs of Table 3, the weight-decay ablations and the synthetic study use a single fixed seed (default 0), and Table 3 states the seed noise of the upstream reference so that the resolvable differences there are explicit. The distinction matters for reading the tables: a difference smaller than the seed spread is not a result, and only the centralized and federated tables measure that spread. Learning-rate selection is performed once, at seed 0, on a validation split disjoint from the test set, and the selected rate is then reused unchanged for every seed.

Step-size schedules. Each experiment inherits the schedule conventional to its domain, and none of them is the constant rate our rates are proved for. The ResNet runs anneal cosinally to zero, the standard for this architecture and the premise of every accuracy we can be compared against; the nanoGPT runs keep record #40's stable-then-decay schedule, flat for the first 55% of steps and linear to $0.1\eta_0$ thereafter (Appendix A.16), because altering it would forfeit the reproduction that validates our port. The discrepancy is deliberate: matching the analysed step size would sacrifice comparability on both benchmarks and close only one of several gaps between the theory and the runs, the others being Newton–Schulz in place of an exact oracle, momentum, and normalization layers. One measurement is exempt: the growth-exponent diagnostic of Appendix A.17, which is run at a constant rate because under a decaying one the accumulated update saturates and the fit reports the schedule instead of the alignment.

Learning-rate selection. The only tuned hyperparameter is η_0. Every method with a norm-fixed step (the eight SIGN- and LMO-family rules of Appendix A.17, SignSGD included) is tuned and reported under one per-layer rule, the unit-gain rule (9), so that η_0 means the same thing across them, namely the per-step RMS gain, and no comparison within the families is confounded by the scaling convention. SGD and Adam have no norm-fixed step for the rule to act on and are run as practitioners run them, with one global rate. Because (9) is a heuristic rather than a theorem, the SIGN family is additionally re-tuned under the two competing conventions: one global rate, and μP's $1/\text{fan-in}$. Appendix A.17 reports whether the ranking moves. Momentum is fixed at 0.9 and weight decay at 0 for the primary tables, the setting Mishra, Trivedi, and Kumar (2026)'s own sweep selects; the regularized case is an ablation (Appendix A.13 centralized, and below for the federated study). The auxiliary group — biases, normalization parameters, the classifier head — is AdamW at a fixed 10^{-3} for every method, so the matrix rule is the only thing that varies.

In the centralized study, selection uses a fixed $45\text{k}/5\text{k}$ train/validation partition and validation accuracy alone, averaged over the last five epochs rather than read off a single one; **the test set is never consulted during tuning**. Each method receives the same number of configurations, drawn from a 1–2–5 lattice (three points per decade), so that tuned rates are exactly reproducible from the tables and every method searches the same grid. An optimum at an endpoint is treated as a failure rather than a result: the grid is widened along the lattice and the method re-run, up to four times. The selected η_0 is then retrained on the full 50k training set for 75 epochs under a cosine schedule, at three seeds, and we report the mean and standard deviation of the test accuracy over the last five epochs. Differences smaller than the seed spread are not claimed as results.

Tuning the federated study. Comparing sign placements is informative only if no placement is handicapped by its step size, so every method receives the same tuning budget: a five-point 1–2–5 lattice in η_0, ranked at a 400-round proxy horizon on a 5k validation split held out of the 50k *before* the client partition, with the grid widened and the method re-tuned whenever an optimum occurred at an endpoint. The reported runs then use the full 50k at the selected rate, so no test image is ever scored during selection. Two assumptions this protocol makes are checked rather than asserted: that the short proxy ranks rates the same way the full horizon does (re-running the top two rates of SignMuon and EF21-MuonSign at 2000 rounds returns the same winner in both cases), and that the shape dependence of the step is carried by the per-layer multiplier rather than by η_0. The second assumption is borne out: seven of the nine methods carrying a per-layer multiplier (the eight of Appendix A.17 plus the server-side-LMO Muon control) select the identical $\eta_0 = 0.05$ across both families, the remaining two (Muon at 0.1, SignSGD at 0.02) selecting a value one lattice step away. Weight decay is 0 in the reported table, matching the setting the theorems analyse; switching on a decoupled 5×10^{-4} moves the top three methods by at most 0.32 points and reorders nothing.

Accuracy and the threshold column. Alongside final accuracy we report the number of epochs (rounds, in the federated tables) to reach a fixed test accuracy. The two measure different things: with the methods spanning about a point and a half at 75 epochs (Table A7), final accuracy is close to the noise floor, whereas a threshold crossing on a monotonically rising curve separates the methods by factors rather than by tenths of a point, and is the analogue of the "steps to 3.35" column of Table 3.

Conventions with numerical consequences. Three implementation conventions can move reported numbers and are recorded here. (i) The Muon LMO is computed in `bfloat16` (five Newton–Schulz steps) unless stated otherwise; for the methods that sign the LMO *output*, entries of $\mathrm{polar}(\cdot)$ near zero can flip at this precision, so their trajectories carry a precision-dependent component (`-lmo-dtype float32` is available). (ii) In the federated runs, BatchNorm running statistics are never updated: local models are discarded each round and BN runs in inference mode during gradient accumulation, so the statistics stay at their initialization for the entire run, in training and evaluation alike. The result is a fixed normalization with learnable affine parameters, self-consistent between train and test, applied identically to every method. It is also one reason a channel can be inactive across an entire local batch and so contribute an exactly zero row to the momentum. (iii) For EF21-MuonSign, training metrics are logged at the broadcast model $\mathbf{W}$ (where gradients must be evaluated) while validation and test metrics are evaluated at the server model $\mathbf{X}$ of (17), the iterate the guarantee bounds, except where a table states otherwise.

A.11 The smooth convex problem

Before the network experiments we measure, on a deterministic L-smooth convex quadratic, the scalar every guarantee in this paper rests on: the alignment $\rho_t = \langle \nabla F(\mathbf{X}_t), \mathbf{D}_t \rangle / (\|\nabla F(\mathbf{X}_t)\|_F \|\mathbf{D}_t\|_F)$ between the gradient and the step taken. Theorems 1–3 construct instances that drive it negative; on random instances it stays positive throughout **[fill]**, so the failure they exhibit is not one random data produces. The same experiment shows what sign compression of an LMO step actually yields: a lower accuracy floor rather than a faster rate **[fill]**. At a matched step size SignSGD descends faster, but its floor is the higher of the two, and it is the floor that a fixed-target iteration count ranks. Section A.12 retains the earlier fixed-target comparison.

Construction. To isolate the effect of matrix structure from stochastic noise and from the complexity of DNN architectures, we use a deterministic L-smooth convex quadratic:

$$F(\mathbf{X}) = \frac{1}{2}\|\mathbf{A}^{1/2}\mathbf{X}\mathbf{B}^{1/2}\|_F^2 = \frac{1}{2}\langle \mathbf{X}, \mathbf{A}\mathbf{X}\mathbf{B} \rangle \to \min_{\mathbf{X} \in \mathbb{R}^{m \times n}}, \quad (30)$$

with $\mathbf{A} \in \mathbb{S}^m_{++}$, $\mathbf{B} \in \mathbb{S}^n_{++}$ symmetric and $\mathbf{X}_0$ drawn entrywise from $\mathcal{N}(0, 0.01)$. Eigenvalues are sampled uniformly from $(0, 1)$ in a Haar-random eigenbasis, so the matrices are almost surely positive *definite* and the minimizer is $\mathbf{X}^\star = \mathbf{0}$ with $F^\star = 0$.

Algorithm	$\min_t \rho_t$	**median** ρ_t	**mean** ρ_t	**% of steps with** $\rho_t < 0$	**closed form**	**tuned** (η, μ)
to be filled: nine rows from the alignment section of `results/synthetic/SUMMARY.md`						

Table A3: Gradient/step alignment ρ_t of Equation (32) along the tuned trajectory, one row per method. The closed-form column is evaluated at $\mathbf{X}_0$ without momentum, so it is comparable only to rows whose tuned μ is 0; with momentum the step is built from $\mathbf{M}_t$ rather than the current gradient and ρ_t falls accordingly. The three sign-around-the-LMO methods admit no closed form; SGD's is 1.

Two facts about this instance are exact rather than estimated, and both are used below. The Hessian of F is the Kronecker product $\mathbf{B} \otimes \mathbf{A}$, so its eigenvalues are the products $\lambda_i(\mathbf{A})\lambda_j(\mathbf{B})$: the Frobenius smoothness constant is $L = \max_{ij} \lambda_i(\mathbf{A})\lambda_j(\mathbf{B}) \leq 1$ and the strong-convexity constant is $\sigma = \min_{ij} \lambda_i(\mathbf{A})\lambda_j(\mathbf{B}) > 0$. The uniform draw leaves the resulting condition number L/σ uncontrolled (it is near 2.5×10^5 at $m = n = 500$), so where conditioning is the variable we instead use log-spaced spectra with $L = 1$ and L/σ set exactly. And since $\nabla F(\mathbf{X}) = \mathbf{AXB}$ in closed form, the gradient can be evaluated at any point without an autograd graph, which is what lets the bidirectional method be scored on its exact model $\mathbf{X}$ while its gradient is taken at the broadcast model $\mathbf{W}$, as its algorithm requires.

What "iterations to a target" measures. The natural criterion — fewest iterations to $F(\mathbf{X}) \leq 10^{-3}$ within $T_{\max} = 5000$, learning rate and momentum tuned per method — turns out not to measure a convergence rate, and reporting it as one is misleading. Every method here except SGD takes a *normalized* step: $\|\operatorname{sign}(\cdot)\|_F = \sqrt{mn}$ and $\|\operatorname{polar}(\cdot)\|_F = \sqrt{r}$ whatever the gradient does. A constant η therefore cannot converge; the iterate settles into a ball of radius $\eta\|\mathbf{S}\|_F$ and F plateaus. Measured directly, the plateau obeys $\|\nabla F\|_\infty \propto \eta$ and $F_\infty \propto \eta^2$, and the iteration count is const$/\eta$. The tuner thus returns the largest η whose plateau still fits under the target, and the ranking it produces is a ranking of *accuracy floors*. The two effects can disagree: at matched η, SignSGD descends faster per step than SignMuon, while SignMuon's floor is the lower of the two **[fill both from `results/synthetic/SUMMARY.md`, floor section]**. We therefore report the floor and the rate separately, and read the descent lemma

$$F(\mathbf{X}_{t+1}) \leq F(\mathbf{X}_t) - \eta\langle \nabla F(\mathbf{X}_t), \mathbf{D}_t\rangle + \tfrac{\eta^2 L}{2}\|\mathbf{D}_t\|_F^2 \tag{31}$$

as the statement that separates them: the second term is the floor, the first is the rate.

Alignment. Equation (31) makes progress contingent on a single scalar, the normalized alignment between the gradient and the step actually taken,

$$\rho_t := \frac{\langle \nabla F(\mathbf{X}_t), \mathbf{D}_t\rangle}{\|\nabla F(\mathbf{X}_t)\|_F \, \|\mathbf{D}_t\|_F} \in [-1, 1]. \tag{32}$$

Theorems 1–3 are constructions that drive ρ_t negative. Table A3 reports its distribution along the tuned trajectory on random instances, which is the empirical counterpart of those theorems and the one measurement here that is about the methods rather than about the tuning protocol. Three closed forms anchor it: $\rho = 1$ for SGD, $\rho = \|\mathbf{G}\|_1/(\|\mathbf{G}\|_F\sqrt{mn})$ for SignSGD, and $\rho = \|\mathbf{G}\|_*/(\|\mathbf{G}\|_F\sqrt{r})$ for Muon. The three sign-around-the-LMO methods admit none, which is the subject of this paper.

Floor, budget and stability. Three further measurements, each testing a quantity the theory predicts in closed form. (i) The floor: the fitted slope of $\log\|\nabla F\|_\infty$ against $\log\eta$, against the 1 that Equation (31) predicts, with coefficient $L\|\mathbf{S}\|_F/2\rho$. Balancing the two terms of (31) gives $\eta\rho\|\nabla F\|_F\|\mathbf{D}\|_F = \eta^2 L\|\mathbf{D}\|_F^2/2$. Since SignMuon and SignSGD share $\|\mathbf{S}\|_F = \sqrt{mn}$ exactly, any gap between their floors is attributable to ρ alone. (ii) The budget exponent: tuning $(\eta, \mu, \text{schedule})$ separately at each horizon T and fitting err $\propto T^{-p}$, $\eta^\star \propto T^{-q}$, where err $:= \min_{t \leq T}\|\nabla F(\mathbf{X}_t)\|_*^2$ is the *squared* dual norm our theorems bound. Reporting the norm itself halves p and leaves q unchanged: q describes the tuned step size, which does not know which power of the error we chose to plot. The nonconvex bound our theorems prove gives $p = q = 1/2$; a strongly convex problem gives $p = q = 1$, because the rate term contracts geometrically and the error is floor-limited. The fit therefore reports which regime the instance is in, and this one, having $\sigma > 0$, need not be the regime the theorems describe. The schedule is tuned per method rather than imposed, since nothing requires the eight methods to prefer the same one. (iii) The stability edge: the largest stable η, with SGD as a control, since its value must reproduce the textbook $2/L$. Reported as a step length $\eta_{\max}\|\mathbf{S}\|_F$, this would be family-independent if the operative trust region were the Frobenius ball; the spread measures how far the Frobenius bound is from the geometry each step actually lives in.

Conditioning. Conditioning is the only knob that governs the dynamics of a quadratic, and the construction above leaves it to chance. Figure A3 sweeps it over decades at fixed $L = 1$.

Protocol and reproduction. Learning rates are searched on logarithmic grids spanning three to four decades. The one-decade linear grids used for the earlier results below were narrow enough to leave optima censored at a grid boundary, which is not a tuned value but an upper bound; the search now flags any optimum occurring at an edge. Every number in this subsection is produced by `python3 -m synthetic.run_gpu`, which writes each stage to `results/synthetic/` together with the commit, GPU and wall time it ran under.

	floor		budget				stability
Algorithm	$\mathrm{d}\log\|\nabla F\|_\infty/\mathrm{d}\log\eta$	R^2	p	R^2	q	R^2	$\eta_{\max}\|\mathbf{S}\|_F$
to be filled: ten rows from the floor, budget-scaling and stability sections of `results/synthetic/SUMMARY.md`							
predicted	1		1/2 or 1		1/2 or 1		$2/L$

Table A4: Floor exponent, budget exponents and stability edge, one row per method, against the values the theory predicts. SGD has no floor (its step vanishes with the gradient), so its floor columns are empty by construction, and its $\eta_{\max}$ is the control that must reproduce $2/L$.

to be generated: the accuracy floor of a constant step

(a) Floor $\|\nabla F\|_\infty$ vs. η

to be generated: best error at each tuned budget

(b) Error vs. budget T

Figure A2: Log–log fits behind Table A4. Left: the accuracy floor of a constant step. Right: best error at each tuned budget.

to be generated: best $\|\nabla F\|$ against the condition number, tuned at each point

Figure A3: Best $\|\nabla F\|$ within the budget against the condition number L/σ, tuned at each point.

A.12 Superseded results for the smooth convex problem

The tables and figure below are the earlier version of this experiment, retained for comparison. They use the protocol criticized above (fewest iterations to $F \leq 10^{-3}$ at $m = n = 500$, one-decade linear grids, a single instance), so they rank accuracy floors rather than convergence rates. Two rows are additionally not tuned values: SGD's $(\eta, \mu) = (0.1, 0.95)$ is the upper corner of both of its grids, and the two EF21 optima lie below the grid Table A6 states for them.

The results of the grid search with the number of iterations required to converge to 0.001, are summarized in Table A5. Muon converges in 1122 iterations, comparable to SGD (972) but slower than Adam (202). SignMuon reaches the target in 2104 iterations, outperforming SignSGD (3120). EF21-MuonUSign requires 3694 iterations and EF21-MuonSign 3564; their overhead on this benign convex problem is expected, since their purpose is to guarantee convergence on problems where SignMuon does not (Theorem 1). The optimization trajectories are shown in Figure A4.

Table A6 details the grid-search ranges swept for each optimizer, and Figure A4 shows the loss and gradient-norm dynamics.

A.13 Centralized training results

Table A7 reports the centralized CIFAR-10 results on ResNet-18 in full: the selected η_0, the final train and test accuracies, the epoch count to the 90% threshold, and the per-epoch cost. Figure 2 starts at epoch 25, where the methods are a few points apart rather than twenty. Every run uses batch 128, momentum 0.9, auxiliary rate 10^{-3}, and zero weight decay, so the matrix rule and η_0 are the only things that vary. Figure A5 gives the accuracy curves from epoch 1, Figure A6 the training loss, and Figure A7 the learning-rate sweep behind the selection.

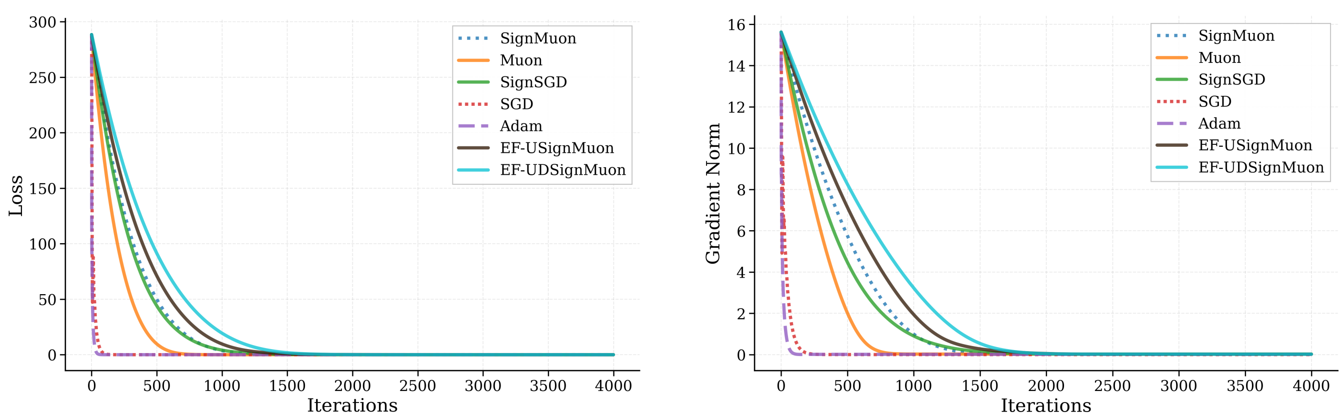
Horizon stability. Selection was carried out at 15 epochs and the reported runs are 75, so the tuning horizon and the reporting horizon differ. Re-tuning the top three rates directly at 75 epochs moves the winner for two methods: SignMuon from $\eta_0 = 0.02$

Algorithm	Learning rate (η)	Momentum (μ)	Iterations to 0.001
SignMuon	0.0002	0.2	2104
MuonUSign	-	-	-
EF21-MuonUSign	0.0033	0.1	3694
EF21-MuonSign	0.0028	0.1	3564
Muon	0.0065	0.1	1122
SignSGD	0.00015	0.8	3120
SGD	0.1	0.95	972
Adam	0.07	-	202

Table A5: Tuned learning rates and momentum coefficients in the experiments on the smooth convex optimization problem (Equation 30).

Algorithm	Learning Rate (η)	Momentum (μ)
SignMuon	$[1 \cdot 10^{-4}, 1 \cdot 10^{-3}]$ with step $1 \cdot 10^{-4}$	$\{0.1, 0.2, \ldots, 0.9, 0.95\}$
MuonUSign	$[1 \cdot 10^{-4}, 1 \cdot 10^{-3}]$ with step $1 \cdot 10^{-4}$	$\{0.1, 0.2, \ldots, 0.9, 0.95\}$
EF21-MuonUSign	$[5 \cdot 10^{-3}, 2 \cdot 10^{-2}]$ with step $1 \cdot 10^{-4}$	$\{0.1, 0.2, \ldots, 0.9, 0.95\}$
EF21-MuonSign	$[5 \cdot 10^{-3}, 2 \cdot 10^{-2}]$ with step $1 \cdot 10^{-4}$	$\{0.1, 0.2, \ldots, 0.9, 0.95\}$
Muon	$[5 \cdot 10^{-3}, 2 \cdot 10^{-2}]$ with step $1 \cdot 10^{-3}$	$\{0.1, 0.2, \ldots, 0.9, 0.95\}$
SignSGD	$[5 \cdot 10^{-5}, 2 \cdot 10^{-4}]$ with step $1 \cdot 10^{-5}$	$\{0.1, 0.2, \ldots, 0.9, 0.95\}$
SGD	$[1 \cdot 10^{-2}, 1 \cdot 10^{-1}]$ with step $1 \cdot 10^{-2}$	$\{0.1, 0.2, \ldots, 0.9, 0.95\}$
Adam	$[1 \cdot 10^{-2}, 1 \cdot 10^{-1}]$ with step $1 \cdot 10^{-2}$	–

Table A6: Hyperparameter search grid for the synthetic convex problem.



(a) Loss

(b) Frobenius norm of the gradient

Figure A4: Smooth convex problem (Equation 30 for 500×500).

to 0.2 and Muon from 0.05 to 0.01. Both moves are small in accuracy — SignMuon reaches 94.53 ± 0.22 at 0.02 against 94.29 at 0.2 (one seed), and the sweep in Figure A7 is flat across that range for every LMO method — but the short-horizon selection is provisional in the strict sense, and we report it as such rather than presenting the 15-epoch grid as if it settled the 75-epoch question.

Weight decay. The primary table is unregularized, the setting the theorems analyse and the one Mishra, Trivedi, and Kumar (2026)'s sweep selects. Repeating the top three at the same η_0 with decoupled decay 5×10^{-4} (seed 0, decay not re-tuned) leaves the ordering intact: EF21-SignMuon 94.52% against 94.55% undecayed, Muon 94.33% against 94.53%, SignMuon 94.35% against 94.75%. Decay is applied decoupled, $\mathbf{X} *= 1 - \eta\lambda$, so the LMO sees the true gradient; the coupled convention would only rotate the direction, since every step here is scale-invariant.

Dataset	Optimizer	Epochs	η_0	Train Acc	Test Acc	Ep. to 90%	s/epoch
CIFAR-10	EF21-SignMuon	75	0.01	100.00	94.64 ± 0.24	8.7	14.8
	SignMuon		0.02	100.00	94.53 ± 0.22	8.3	14.4
	Muon		0.05	99.99	94.49 ± 0.09	7.7	14.4
	EF21-MuonSign		0.01	99.99	94.22 ± 0.13	10.3	15.3
	EF21-MuonUSign		0.2	99.99	94.15 ± 0.16	8.7	14.9
	MuonUSign		0.02	99.99	93.90 ± 0.08	10.0	14.4
	MuonSign		0.005	99.98	93.65 ± 0.21	15.0	14.5
	Adam		0.0005	99.98	93.35 ± 0.15	18.8	12.7
	SignSGD		0.002	99.97	93.27 ± 0.16	19.0	12.1
	SGD		0.02	99.95	93.13 ± 0.07	21.7	12.2

Table A7: CIFAR-10: centralized learning on ResNet-18, three seeds. Test accuracy is the mean $\pm$ standard deviation of the last five epochs; train accuracy, "Ep. to 90%" and "s/epoch" are means over the same three seeds. η_0 was selected on a held-out 5k validation split at 15 epochs and then retrained on the full 50k set, as described in Appendix A.10. Every method fits the training set to within 0.05 points of the others, so the column separates nothing on its own; the differences in the test column are not differences in how far training got.

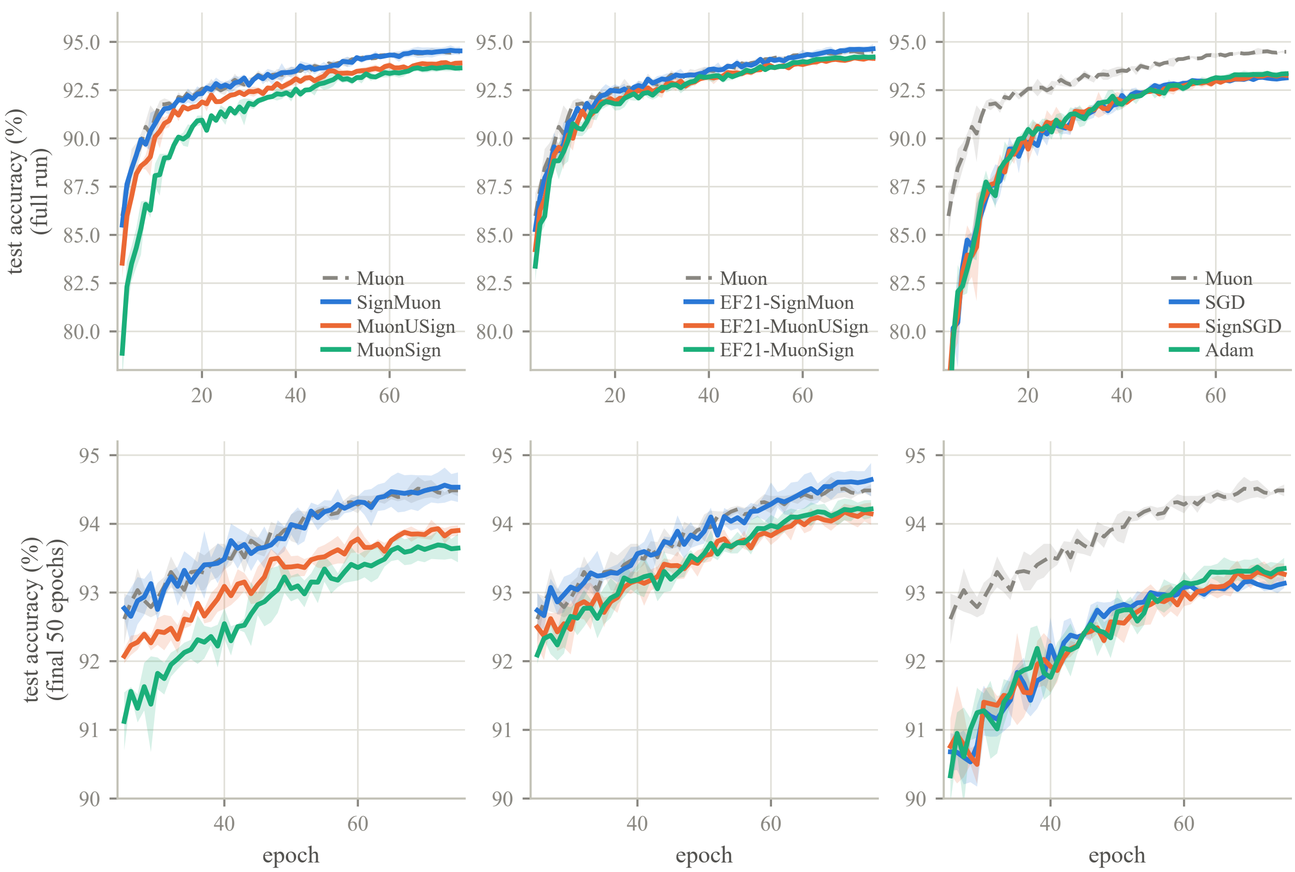


Figure A5: Test accuracy over the whole run (top row) and from epoch 25 (bottom row), the same data at two scales, grouped as in Figure 2. Bands are ± 1 standard deviation over three seeds; Muon is the gray dashed reference in every panel. Series are labelled at the right of each panel.

A.14 Communication accounting

Table 2 quotes mean bits per parameter per round. This section says exactly what is counted, because the headline "32×" of the sign-compression literature is an idealization that three separate effects erode, and only one of them is usually mentioned.

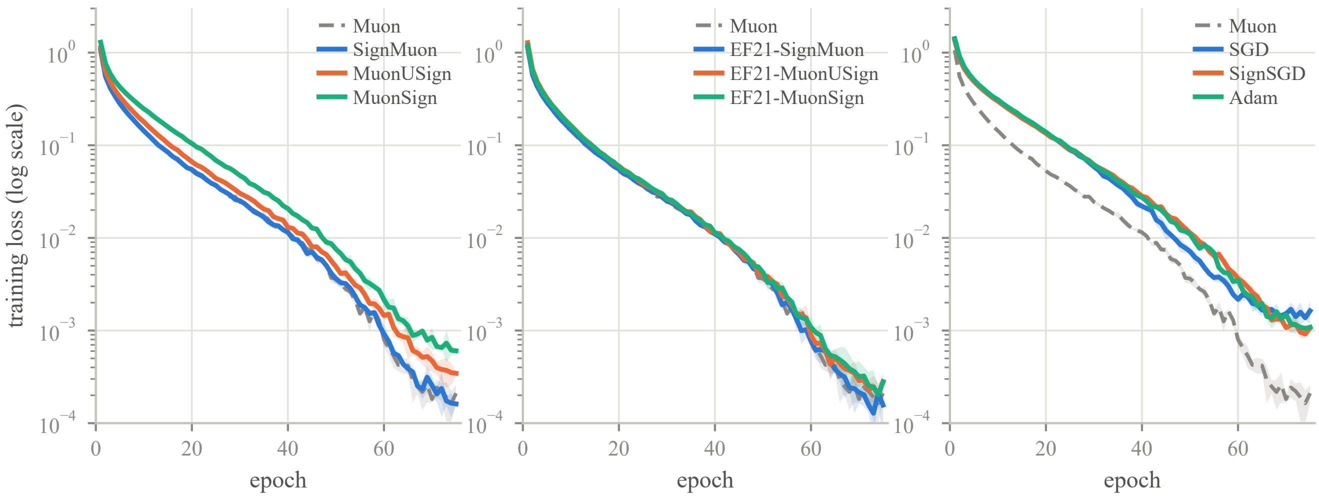


Figure A6: Training loss (log scale), grouped as in Figure 2 and over the same runs. Within a panel the methods are hard to separate (every one of them falls to 10^{-3} or below, and they differ mainly in how early), which is why the body figure carries the accuracy alone. What the loss does show is the gap between families: in the right panel Muon reaches a given loss some fifteen epochs ahead of SGD, SignSGD and Adam and ends nearly an order of magnitude lower, while in the left panel SignMuon tracks Muon and the sign-before and both-sides orderings trail it, in the same order as their accuracies.

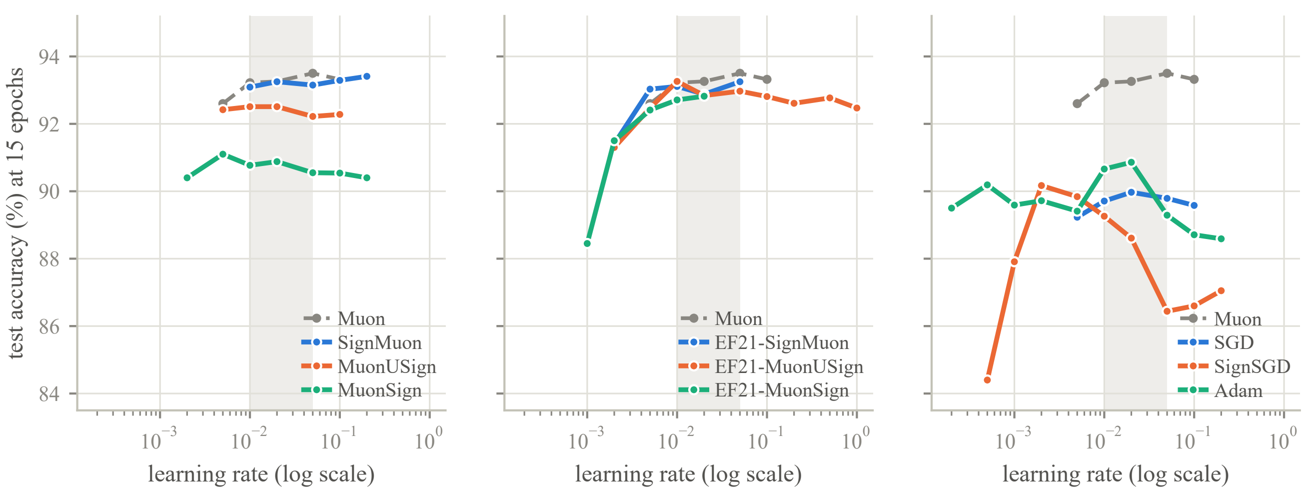


Figure A7: Learning-rate sweep at 15 epochs on the tuning split, grouped as in Figure 2, with Muon repeated as the gray dashed reference. The shaded band marks $\eta_0 \in [0.01, 0.05]$, the range every method except EF21-MuonSign was swept over and therefore the only range-matched comparison: within it the LMO methods vary by 0.16–0.42 points (Muon itself, 0.28, is no flatter than the sign variants), while Adam varies by 1.57 and SignSGD by 2.82. The robustness belongs to the LMO, and taking the sign of its output does not cost it.

Write P_{mat} for the number of matrix parameters, P_{aux} for the auxiliary group (biases, BatchNorm affine parameters, the classifier head), L for the number of matrix layers, and $P = P_{\text{mat}} + P_{\text{aux}}$. On CNN2, $P_{\text{mat}} = 762{,}560$, $P_{\text{aux}} = 2{,}146$ and $L = 3$. All figures below are *per client per round*: the uplink is what one client sends, the downlink what the server sends to one client.

(i) The uplink alphabet. Under the convention of Section 4 every exact zero is randomized to ± 1, so a transmitted symbol is a genuine bit and no entropy coding is required to realize it. Retaining the third symbol instead would make the alphabet ternary at $H\big(p_0, \frac{1-p_0}{2}, \frac{1-p_0}{2}\big)$ bits, which is 1.02 bits at the 0.2% zero rate CNN2 exhibits and 1.37 at a 10% rate. That is close in the first case, but only realizable with a coder, which is the practical argument for randomizing.

Method	Up (bits)	Down (bits)	Up	Down
Muon, MuonServer	32	32	$1.0\times$	$1.0\times$
SGD, Adam	32	32	$1.0\times$	$1.0\times$
SignMuon	1.0870	1.0870	$29.4\times$	$29.4\times$
SignSGD	1.0870	1.0870	$29.4\times$	$29.4\times$
MuonSign	1.0870	1.0870	$29.4\times$	$29.4\times$
EF21-MuonSign	1.0871	1.0871	$29.4\times$	$29.4\times$
MuonUSign	1.0870	32	$29.4\times$	$1.0\times$
EF21-SignMuon	1.0871	32	$29.4\times$	$1.0\times$
EF21-MuonUSign	1.0871	32	$29.4\times$	$1.0\times$

Table A8: Communication per client per round on CNN2, under the randomized-zero convention. Bits are per model parameter; reductions are against a 32-bit baseline. The fourth decimal separates the error-feedback methods from the rest: it is the per-layer scale of point (iii). Groups: uncompressed references, one bit in both directions, one bit on the uplink only.

(ii) The auxiliary group is never compressed. It travels at full precision in both directions for every method, so a "one-bit" channel actually costs

$$\frac{1 \cdot P_{\mathrm{mat}} + 32\, P_{\mathrm{aux}}}{P} = 1.087 \text{ bits},$$

a $29.4\times$ reduction rather than $32\times$. On CNN2 the group is 0.28% of the parameters; on an architecture with a large embedding or head it would dominate this table, which is why the quantity is computed per model rather than quoted.

(iii) Error feedback carries one scale per layer. Both EF21 channels transmit the pair $(\mathbf{s}_t, \alpha_t)$ with one full-precision α_t per matrix layer: on the uplink for every EF21 method, and again on the downlink for EF21-MuonSign. That is $32L$ bits, adding $32L/P \approx 1.3 \times 10^{-4}$ bits per parameter here. That is four decimal places in, and it is reported for completeness rather than because it changes a conclusion. It is the difference between "one bit per parameter" and "one bit per parameter, plus a constant".

(iv) Which methods compress the downlink. The criterion is not whether the method applies a compressor but whether the object the server must distribute is already ± 1-valued. Three cases qualify: the majority vote $\mathbf{s}_t^{\mathrm{agg}}$ itself (SignMuon, SignSGD), a signed LMO output (MuonSign), and a primal error-feedback residual (EF21-MuonSign). In the first of these the server broadcasts the vote rather than the model and each client applies the step to its own replica; replicas start from a common $\mathbf{X}_0$ and receive identical updates, so they never drift. The remaining three methods must distribute a dense server-side quantity — $\mathrm{polar}(\cdot)$ of the aggregate for MuonUSign and EF21-MuonUSign, a scaled average of signs for EF21-SignMuon — and therefore transmit them at full precision.

Computed by `federated.algorithms.communication_bits`, which takes the measured zero rate of the run it describes, so a run made under the legacy ternary convention reports its own higher figure rather than the idealized one.

A.15 Federated training results

Figure A8 (together with Table 2 in the main text) reports the comparison of optimizers at $N = 11$ clients. The figure additionally shows Muon with a server-side LMO, which is not a communication-efficient method but isolates whether moving the oracle off the clients carries a penalty on its own; it does not ($85.74 \pm 0.11\%$ against Muon's $85.99 \pm 0.26\%$), so the gaps in Table 2 are attributable to compression and sign placement rather than to where the oracle runs.

A.16 Language-modelling details

Setup. Upstream modded-nanoGPT record #40 (2025-10-04), the last record before NorMuon and hence the last whose hidden-matrix optimizer is a clean, separable momentum → LMO → step Muon, so our variants inject exactly at the LMO. Model: 12 layers, model dimension 768, 6 heads of dimension 128, vocabulary 50,257; hidden matrices are 768×3072 (the merged $\mathbf{Q}/\mathbf{K}/\mathbf{V}/\mathbf{O}$ weight is used as four 768×768 blocks, and both the LMO and the compressor scale are applied per block). Data: FineWeb10B, 262,144 tokens per step, 2330 steps (= 611M tokens), validation on the fixed 10,485,760-token split. Hardware: 8×H100, one process per GPU; gradients are averaged by `reduce_scatter` so the owning rank runs the centralized update and `all_gather` returns the parameter, i.e. the compression is a property of the update rule, as in the centralized algorithms we analyze.

Hyperparameters. Matrix/gate optimizer: $\eta_0 = 0.06$ (LMO family) or 0.03 (SIGN family), per-layer scaled by the unit-gain rule (Appendix A.17); Nesterov momentum $\mu = 0.95$, warmed up linearly from 0.85 over the first 300 steps and cooled back to 0.85 over the last 50; weight decay 0 (the record's own value); η constant then linearly cooled to $0.1\eta_0$ over the final 45% of the run; LMO by 5 Polar-Express iterations. Auxiliary parameters (embeddings, scalars, head) use the record's distributed Adam unchanged: $\eta = 0.008$, $\beta = (0.65, 0.95)$, $\varepsilon = 10^{-8}$, no weight decay, per-parameter multipliers 75 on embeddings and 5 on

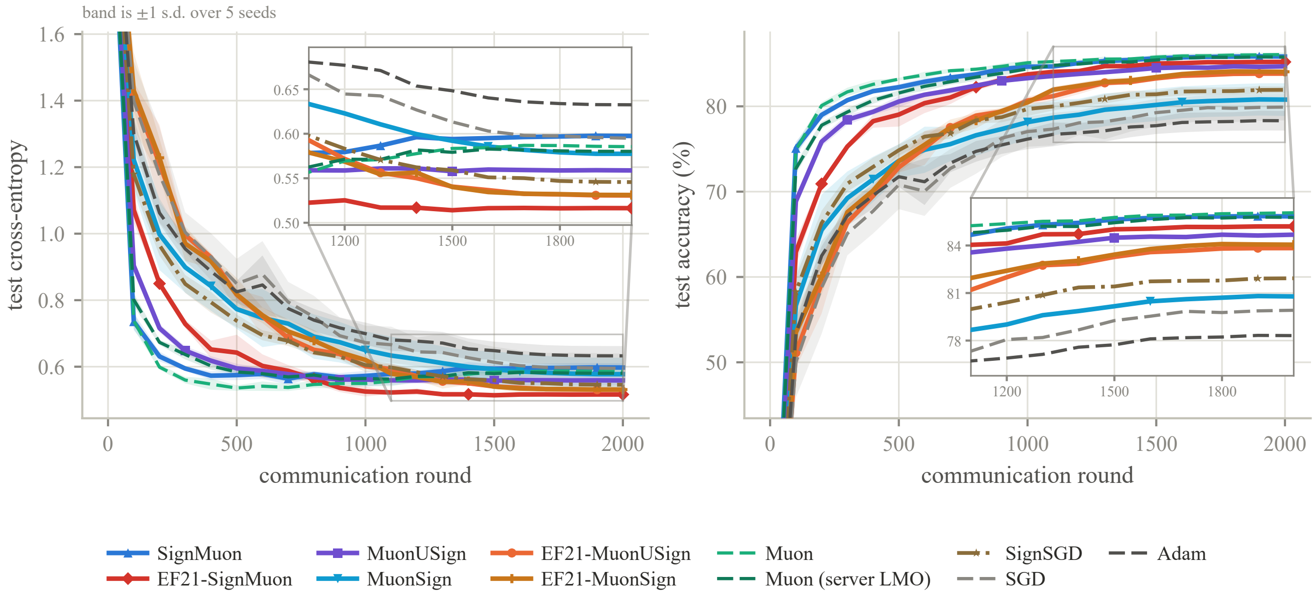


Figure A8: CIFAR-10 federated learning on CNN2 at $N = 11$ clients and batch 192 per client: test cross-entropy (left) and test accuracy (right) against communication round. Each curve is the mean over five seeds and the band is ± 1 sample standard deviation across them; all methods are evaluated on the exact server model $\mathbf{X}_t$. Both axes are clipped to exclude round 0, which is the untrained model and identical for every method. Line style carries the method's role — solid for the one-bit methods, dashed for the uncompressed references, dash-dot for SignSGD — so that hue is not the only channel separating eleven curves; markers are spaced apart per method for the same reason. The insets magnify the final 45% of rounds, where every number in Table 2 is read and where the curves lie within a line width of one another; bands are omitted inside them, since at that zoom they overlap into a single wash. Eleven methods are drawn, one more than Table 2, the extra being the server-side-LMO Muon control.

scalars, stepped every other iteration. Nothing above was tuned by us: the LMO family runs at the record's own η_0, and every value outside the matrix optimizer is the record's. Wall-clock is within 1% across all eight methods (61.4–62.1 ms/step).

Compressor diagnostics. Table A9 reports, per layer type at the final step, the contraction each scaled sign achieves, $\alpha(\mathbf{\Delta}) = \|\mathbf{\Delta}\|_1^2/(d\|\mathbf{\Delta}\|_F^2)$, and the relative estimator lag. Three points deserve separate comment. *(i)* Every uplink is well-contractive, $\alpha \in [0.32, 0.64]$ against the isotropic $2/\pi = 0.637$, and EF21-SignMuon's is uniformly the best because its target is orthogonal; yet the uplink *lag* is large (0.36–0.82), meaning the estimator is far from its target at every step and the methods nonetheless train well – the LMO is scale-invariant and needs only the direction. *(ii)* The single anomaly in the table is the `c_proj` downlink, $\alpha = 1.2 \times 10^{-4}$, four orders of magnitude below its uplink on the very same layer. Since both compressors are the same operator, the difference is a property of the residual, not of the compressor: the uplink residual is refreshed by an exogenous stochastic gradient each round, the downlink residual is generated by the compressor's own recursion (Remark 3). *(iii)* The resulting validation loss at the server model $\mathbf{X}$ is 4.20 through step 750, rises to 5.52 by step 1500, and then holds there (5.51–5.58 to the end of the run): a persistent offset above the broadcast model, not a divergence.

Where the two models part. The `c_proj` anomaly of item *(ii)* is what separates EF21-MuonSign's two models: on that layer the gap $\|\mathbf{X}_t - \mathbf{W}_t\|_F/\|\mathbf{W}_t\|_F$ stops decreasing at 8% against $<1\%$ for every other layer, and that one layer accounts for the entire $\approx$2.2-nat offset of item *(iii)*. The reason is the initialization. A layer built from zero receives maximally correlated updates, so its downlink residual concentrates, and a compressor that moves every coordinate by $\mathrm{mean}\,|\mathbf{\Delta}^{\downarrow}|$ cannot catch the coordinates driven hardest. This is the mechanism of Remark 3 rather than a tuning failure. Lowering η_0 does not repair it, because the admissible step size would have to shrink by a further $\sqrt{r}$ in the layer rank (Remark 4); only a downlink compressor contractive in the *spectral* norm would.

A.17 Per-Layer Step Sizes: the Unit-Gain Rule

This appendix derives the heuristic (9) stated in the main text, selects its one free exponent by measurement, and records what it does and does not deliver (including the two methods it does not act on at all, SGD and Adam, which have no norm-fixed step and are run at one global rate throughout).

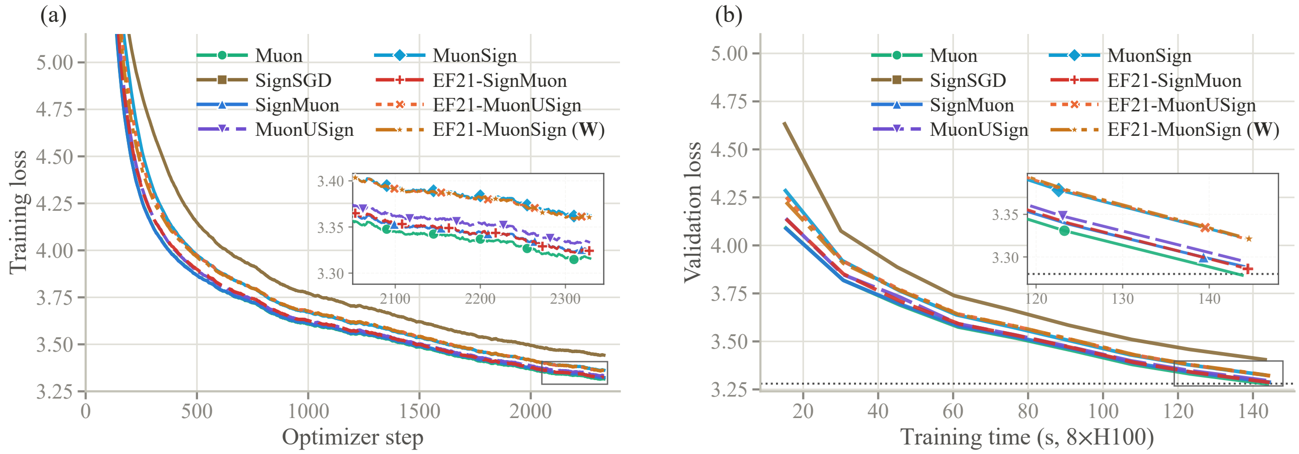


Figure A9: NanoGPT, supporting curves. **(a)** Training loss against optimizer step (EMA-smoothed; the logged quantity is a per-rank sum over 32,768 tokens, divided out here). **(b)** Validation loss against the speedrun clock, which excludes validation and compilation. Insets magnify the boxed tails, as in Figure 3. Marker positions are staggered in both figures so that curves lying within a line width of one another can still be followed individually. The ordering is the same on both axes: no method incurs a measurable wall-clock premium for its compressor or its error-feedback buffers. Note in (a) that EF21-MuonSign's training loss – taken at $\mathbf{W}$ – shows no degradation throughout, which is what isolates its validation gap to the $\mathbf{X}$-tracking rather than to training.

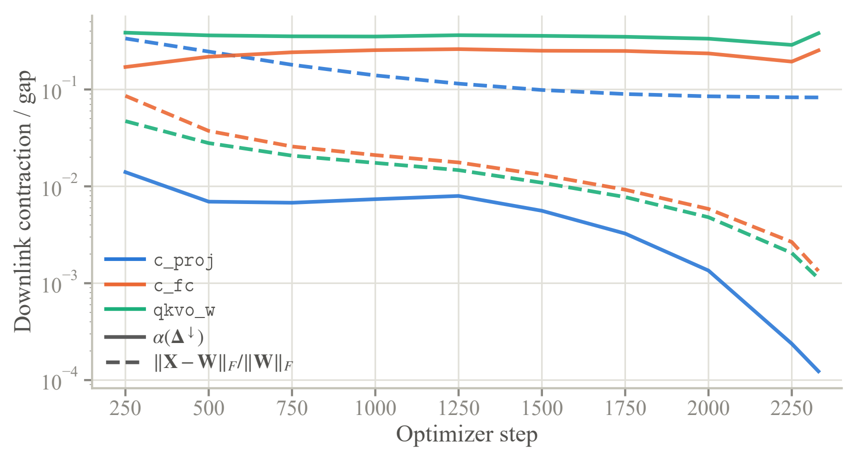


Figure A10: The downlink measurement for EF21-MuonSign, per layer type: the contraction $\alpha(\mathbf{\Delta}^{\downarrow})$ the scaled sign actually achieves (solid) and the resulting server/broadcast gap $\|\mathbf{X}_t - \mathbf{W}_t\|_F/\|\mathbf{W}_t\|_F$ (dashed). On the zero-initialized `c_proj` the contraction is two orders of magnitude below the rest from step 250 on and falls a further two by the end, and the gap stops decreasing near 10^{-1}; every other layer type contracts at $\Theta(10^{-1})$ and closes its gap by 6–64× over the run, ending below 10^{-2}. This is Remark 3 measured.

Our counterexamples, like the analysis of Mishra, Trivedi, and Kumar (2026), concern a single matrix, where one scalar step size suffices. A network has layers of very different shapes, and the methods of this paper produce step matrices from *two* families whose norms scale differently with shape, so a single global η cannot be simultaneously correct for both families and across layers. This is not a gap we must tolerate: the layer-wise LMO framework to which our convergence result reduces (Riabinin et al. 2025; Gruntkowska et al. 2025) already carries per-layer norms $\|\cdot\|_{(\ell)}$, smoothness constants L_ℓ and radii η_ℓ; it is only the *experiments* that have been setting that radius to a constant. What follows instantiates it. The specific rule is this paper's own; the criterion behind it is borrowed (it is the average-case form of the spectral scaling condition of the maximal-update literature (Yang et al. 2021; Yang, Simon, and Bernstein 2023; Large et al. 2024)), and its LMO endpoint reproduces the aspect factor Muon already uses in practice (Jordan et al. 2024), which is the external check we rely on.

Layer type	uplink α	uplink lag	downlink α	downlink gap
EF21-SignMuon				
`qkvo_w` ($\times$10)	0.640	0.66	–	–
`c_fc` ($\times$11)	0.633	0.62	–	–
`c_proj` ($\times$11)	0.634	0.62	–	–
EF21-MuonUSign				
`qkvo_w`	0.374	0.81	–	–
`c_fc`	0.323	0.82	–	–
`c_proj`	0.597	0.62	–	–
EF21-MuonSign				
`qkvo_w`	0.406	0.79	0.380	0.0011
`c_fc`	0.355	0.78	0.253	0.0013
`c_proj`	0.596	0.61	$\mathbf{1.2{\times}10^{-4}}$	$\mathbf{0.083}$

Table A9: Compressor diagnostics at step 2330, medians over the identical layers of each type. α is the contraction the scaled sign achieves on that round's residual ($2/\pi$ for an isotropic residual, $1/d$ in the worst case); "lag" is $\|\text{target} - \text{estimator}\|_F/\|\text{target}\|_F$; "gap" is $\|\mathbf{X}_t - \mathbf{W}_t\|_F/\|\mathbf{W}_t\|_F$. Downlink columns are empty for the two methods that broadcast exactly. Gates are omitted (they are 6×12 and 1×12).

The two families. For a parameter reshaped to $\mathbf{X} \in \mathbb{R}^{m\times n}$ (m = fan-out, n = fan-in, $r = \operatorname{rank} = \min(m,n)$ generically), the step matrices are

$$\mathbf{s} = \mathbf{U}\mathbf{V}^\top \ \ (\textsc{lmo}), \qquad \mathbf{s} \in \{\pm 1\}^{m\times n} \ \ (\textsc{sign}). \tag{33}$$

The LMO family comprises Muon, MuonUSign, EF21-MuonUSign, EF21-MuonSign and EF21-SignMuon; the SIGN family comprises SignMuon, MuonSign and SignSGD. Both have *exactly* known Frobenius norms: $\|\mathbf{U}\mathbf{V}^\top\|_F^2 = \operatorname{tr}(\mathbf{V}\mathbf{U}^\top\mathbf{U}\mathbf{V}^\top) = \operatorname{tr}(\mathbf{V}^\top\mathbf{V}) = r$, so $\|\mathbf{s}\|_F = \sqrt{r}$; and a ± 1 matrix has $\|\mathbf{s}\|_F = \sqrt{mn}$. (EF21-SignMuon steps along the error-feedback estimator $\mathbf{d}_t^{\text{est}}$ of $\operatorname{polar}(\tilde{\mathbf{M}}_t)$ rather than along the oracle output itself, so for it $\|\mathbf{s}\|_F = \sqrt{r}$ holds only in the limit; we assign it to the LMO family on that basis.)

The criterion. Define the *RMS gain* of $\mathbf{A} \in \mathbb{R}^{m\times n}$, i.e. how much it amplifies a generic input in root-mean-square terms, with $\operatorname{rms}(\mathbf{v}) := \|\mathbf{v}\|/\sqrt{\dim}$:

$$\gamma(\mathbf{A})^2 := \frac{\mathbb{E}_{\mathbf{u}}\left[\operatorname{rms}(\mathbf{A}\mathbf{u})^2\right]}{\mathbb{E}_{\mathbf{u}}\left[\operatorname{rms}(\mathbf{u})^2\right]}, \qquad \mathbf{u} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_n). \tag{34}$$

Since $\mathbb{E}\|\mathbf{A}\mathbf{u}\|^2 = \operatorname{tr}(\mathbf{A}^\top\mathbf{A}) = \|\mathbf{A}\|_F^2$ and $\mathbb{E}\|\mathbf{u}\|^2 = n$, (34) evaluates in closed form:

$$\gamma(\mathbf{A}) = \frac{\|\mathbf{A}\|_F}{\sqrt{m}}. \tag{35}$$

No random-matrix theory and no distributional fitting enter; the single modelling assumption is that $\mathbf{u}$ is isotropic *and independent of* $\mathbf{A}$, which we return to below. Controlling a layer update's RMS-to-RMS effect is precisely the desideratum of the spectral scaling condition (Yang, Simon, and Bernstein 2023) and of the modular norm (Large et al. 2024); (34) is its average-case (isotropic-input) version, and when the independence fails — aligned inputs, where the operator norm governs the gain instead — one recovers the μP endpoint $a = 1$ discussed below.

Every standard fan-in-scaled initialization has *shape-independent* gain. For He normal ($\sigma^2 = 2/n$), $\|\mathbf{X}\|_F = \sigma\sqrt{mn} = \sqrt{2m}$, so $\gamma(\mathbf{X}) = \sqrt{2}$; for PyTorch's default Kaiming-uniform convolution ($\sigma^2 = 1/(3n)$), $\gamma(\mathbf{X}) = 1/\sqrt{3}$. Either way a constant. Requiring the update's gain to be a fixed fraction of the weight's is therefore simply the requirement that *the per-step gain be the same on every layer*, and by (35) that is one formula:

$$\boxed{\eta_\ell = \eta_0\,\lambda_\ell, \qquad \lambda_\ell = \frac{\sqrt{m}}{\|\mathbf{s}\|_F}} \tag{36}$$

which gives $\gamma(\eta_\ell \mathbf{s}) = \eta_0$ exactly, for every shape and both families – so η_0 *is* the per-step RMS gain. One caveat: the rule is derived from $\|\mathbf{U}\mathbf{V}^\top\|_F = \sqrt{r}$, which holds for the exact oracle. The five-step Newton–Schulz iteration we actually run returns a step 6–22% shorter than $\sqrt{\min(m,n)}$, so for the LMO-terminated methods the realized gain is flat only up to that factor, and η_0 is the per-step RMS gain of the exact step rather than of the executed one. Substituting the two Frobenius norms of (33),

$$\lambda_\ell^{\text{LMO}} = \sqrt{\max\!\left(1, \tfrac{m}{n}\right)}, \qquad \lambda_\ell^{\text{SIGN}} = \frac{1}{\sqrt{n}} = \frac{1}{\sqrt{\text{fan-in}}}. \tag{37}$$

Why we believe (37). The first expression is exactly the aspect-ratio factor $\sqrt{\max(1, m/n)}$ present in the reference Muon implementation (Jordan et al. 2024), which was introduced as a practical heuristic. The unit-gain criterion *derives* it, and also explains why Muon's step size is known to transfer across widths: its step has $\gamma = \eta$ independently of n, so no fan-in correction is needed. The second expression is the counterpart the sign family has never been given. Only η_0 is tuned, and it is now a shape-free quantity; the shape dependence is fixed a priori.

Two consequences are worth recording. First, λ_ℓ is a deterministic function of the layer shape, known to server and clients alike, so per-layer step sizes require *no* communication and leave the one-bit-per-parameter budget untouched. Second, on the CIFAR ResNet-18 of our experiments $\lambda_\ell^{\text{SIGN}}$ spans a factor of 13 across layers (from $1/\sqrt{27}$ at the stem to $1/\sqrt{4608}$ in the last stage), so a single global rate is necessarily a compromise: roughly correct for the middle of the network, several-fold too large at the stem and too small at the end. The LMO family is spared this, which is one reason full-precision Muon is easier to tune than its sign-compressed variants.

The one open exponent. Writing $\lambda_\ell^{\text{SIGN}} = n^{-a}$, the unit-gain rule is $a = \frac{1}{2}$. Identity (35) requires the input to be independent of $\mathbf{A}$, which is the right model for a *single* step; if the *accumulated* update $\sum_t \eta \mathbf{s}_t$ correlates with the activations, its gain can be as large as $\|\mathbf{A}\|_{\text{op}}\sqrt{n/m} = \Theta(\eta n)$ rather than $\Theta(\eta\sqrt{n})$, giving $a = 1$, the classical μP rule $\eta \propto 1/\text{fan-in}$ for sign-like updates (Yang et al. 2021). The endpoint is thus an empirical question, and one a direct measurement settles: we track the realized gain $\|\mathbf{X}_t - \mathbf{X}_0\|_F/\sqrt{m}$ over training and fit its growth exponent h in t: $h = \frac{1}{2}$ for incoherent (random-walk) accumulation, $h = 1$ for aligned accumulation. The fit must be taken at a constant step size. Under a decaying schedule the accumulation saturates and the fit reports the schedule instead of the alignment; on our cosine-annealed runs it returns $|\hat{h}| < 0.06$ at $R^2 \leq 0.31$, which is not a shallow power law but the absence of one. At a constant rate over 20 epochs, $\hat{h} = 0.513$ for Muon and 0.516 for SignMuon, both at $R^2 = 1.000$.

Muon serves as the control. For the LMO family (37) and the μP rule prescribe the identical multiplier, so Muon's exponent is what the diagnostic returns when the rule is not in question; SignMuon agreeing with it to 0.003 places the sign family's accumulation in the same regime. The measured value is that of the incoherent model, so $a = \frac{1}{2}$, and we adopt it for both families in every network experiment (the synthetic problem of Appendix A.11 has a single matrix parameter, where the multiplier is one constant absorbed into the tuned η). What the exponent fixes is the scaling of the accumulated update in t, hence the shape dependence of λ_ℓ and the transfer of η_0 across widths; it makes no claim about which exponent maximizes accuracy at one fixed width, where a alters only the relative rates across layers and is largely absorbed into η_0.

The placement of weight decay is not arbitrary. The same scale invariance that makes λ_ℓ necessary also dictates *where* an ℓ_2 penalty may be applied. Every step map considered here is positively homogeneous of degree *zero*: $\text{sign}(c\mathbf{M}) = \text{sign}(\mathbf{M})$ and $\text{polar}(c\mathbf{M}) = \text{polar}(\mathbf{M})$ for all $c > 0$. Consequently, folding a decay term into the gradient before the LMO – $\widetilde{\mathbf{G}}_t = \mathbf{G}_t + \lambda_{\text{wd}}\mathbf{W}_t$, the convention of Mishra, Trivedi, and Kumar (2026) and of most sign-method implementations – supplies no contraction at all: the step length $\eta_t\lambda_\ell\|\mathbf{s}\|_F$ is fixed by (33) and the decay term cannot change it, so whatever shrinkage occurs is a by-product of *rotating* the direction rather than the geometric decay the penalty is meant to impose. For the sign-terminated steps the cancellation is exact to machine precision; for the LMO-terminated ones it is exact for the polar factor itself, and the few percent by which a Newton–Schulz approximation of it moves is the iteration's error, not shrinkage. The rotation is governed by the ratio $\rho_t = \lambda_{\text{wd}}\|\mathbf{W}_t\|_F/\|\mathbf{G}_t\|_F$, and it is not benign. Since $\|\mathbf{G}_t\|_F$ falls by orders of magnitude over training while $\|\mathbf{W}_t\|_F$ grows, ρ_t drifts from negligible to $\Theta(1)$; worse for a comparison, it depends on each method's own momentum scale, so one nominal λ_{wd} is a different perturbation for each. Decoupled decay, $\mathbf{W}_{t+1} = (1 - \eta_t\lambda_{\text{wd}})\mathbf{W}_t - \eta_t\lambda_\ell\mathbf{s}_t$, is by contrast exactly commensurate with the update under the unit-gain rule: the displacement it contributes has gain $\eta_t\lambda_{\text{wd}}\gamma(\mathbf{W}_t)$ against the step's gain η_t, so their ratio is $\lambda_{\text{wd}}\gamma(\mathbf{W}_t)$ – free of η_t, of the layer shape, and of the method. We therefore decouple, and use the coupled form only for an ablation. This also explains an otherwise puzzling observation of Mishra, Trivedi, and Kumar (2026): sweeping $\lambda_{\text{wd}} \in \{0, 0.1, 0.2\}$ coupled over 330 CIFAR-10 ResNet-50 configurations, every one of the five Muon and Sign-Muon entries in their top ten (the only entries for which decay was swept at all) selects $\lambda_{\text{wd}} = 0$. At their two nonzero values $\rho_t \gg 1$ throughout training, so the update degenerates towards $-\,\text{sign}(\mathbf{W}_t)$; the sweep rejects the placement rather than the regularizer.

What the analysis covers. Our theorems are stated for unregularized f, so we report unregularized runs as the primary comparison and weight decay as an ablation; the reference nanoGPT configuration we build on also uses $\lambda_{\text{wd}} = 0$ for every parameter group. Two remarks delimit the gap. First, the *coupled* form is covered verbatim: it is nothing other than running the same method on $f_\lambda = f + \frac{\lambda_{\text{wd}}}{2}\|\mathbf{W}\|_F^2$, so every rate carries over with $L_i \mapsto L_i + \lambda_{\text{wd}} r_i$. The rank factor is not slack: our smoothness is measured in the nuclear norm against a spectral-norm displacement (Assumption 2), and $\|\lambda_{\text{wd}}\mathbf{Z}\|_* \leq \lambda_{\text{wd}} r_i \|\mathbf{Z}\|_{2\to 2}$ is tight at $\mathbf{Z} = \mathbf{I}_{r_i}$; $L \mapsto L + \lambda_{\text{wd}}$ would be the Euclidean statement, and these rates are not Euclidean. The irony is that this is exactly the variant that does not regularize. Second, the *decoupled* form is not covered by our rates, but it provides something the analysis wants. Since $\|\mathbf{s}_t\|_F$ is a known constant and $\lambda_\ell\|\mathbf{s}_t\|_F = \sqrt{m}$ identically under (36), the triangle inequality gives $\|\mathbf{W}_{t+1}\|_F \leq (1 - \eta_t\lambda_{\text{wd}})\|\mathbf{W}_t\|_F + \eta_t\sqrt{m}$, and hence, for any $\eta_t\lambda_{\text{wd}} \leq 1$,

$$\gamma(\mathbf{W}_t) \;\leq\; \max\{\gamma(\mathbf{W}_0),\; \lambda_{\text{wd}}^{-1}\} \quad \text{for all } t, \tag{38}$$

a bound on the layer's gain that is uniform in t and *independent of the layer shape*. Norm-constrained updates are what make this possible: for SGD the step length is data-dependent and no such a priori bound exists. Since layer-wise LMO analyses assume

Method	**Rule**	$\eta_0^\star$	**Test acc (%)**
to be filled: three rules $\times$ *the sign family, from* `results/federated/scaling_compare.csv`			

Table A10: Per-layer rule ablation on federated CNN2: each SIGN-family method re-tuned under one global rate, the unit-gain rule (9), and μP. The quantity of interest is the ordering within a column, not the level across columns.

Rule	**LMO**	**SIGN**	**Equalizes**
global ($a\!=\!0$)	1	1	nothing
Muon default	$\sqrt{\max(1, \frac{m}{n})}$	1	LMO gain
unit gain	$\sqrt{\max(1, \frac{m}{n})}$	$n^{-1/2}$	per-step gain, both
μP ($a\!=\!1$)	$\sqrt{\max(1, \frac{m}{n})}$	n^{-1}	aligned accumulation
Mishra et al.	$r^{-1/2}$	$(mn)^{-1/2}$	$\|\mathbf{s}\|_F$

Table A11: Per-layer step-size multipliers λ_ℓ. Only η_0 is tuned; λ_ℓ is fixed a priori by the shape. The rules differ only in the SIGN family, and the LMO column of the unit-gain rule coincides with the factor already used in practice (Jordan et al. 2024) – our main evidence that (34) is the right criterion. The last row is the normalization of Mishra, Trivedi, and Kumar (2026), which equalizes $\|\mathbf{s}\|_F$ rather than the gain.

smoothness on a bounded region, (38) is the statement that decoupled decay supplies that region rather than complicating it. A convergence rate for the decoupled variant itself we do not claim.

Sensitivity: does the rule change any conclusion? Equation (9) is a heuristic, so the honest test is whether our conclusions depend on it. Every SIGN-family method is re-tuned from scratch — same lattice, same validation protocol, same budget — under three conventions: one global rate ($\lambda_\ell = 1$), unit gain ($\lambda_\ell = 1/\sqrt{\text{fan-in}}$), and μP ($\lambda_\ell = 1/\text{fan-in}$). Each rule shifts the *selected* η_0 by roughly the multiplier it prescribes, which is what it is for; what matters is whether it shifts the *ordering* of the methods, and Table A10 reports that it does not **[fill]**. We run this on the federated CNN2 study rather than on ResNet-18 deliberately: CNN2's three matrix parameters span a factor of 7.8 in $\lambda_\ell^{\text{SIGN}}$ with no single shape dominating the parameter count, so a wrong exponent has more scope to become visible there. The LMO family is not swept, since unit gain and μP prescribe the identical multiplier for it.

Comparison with concurrent work. Mishra, Trivedi, and Kumar (2026) analyse the normalized update $\mathbf{D}_t = \bar{\mathbf{S}}_t/\sqrt{mn}$, justified by $\|\mathbf{D}_t\|_{\text{op}} \le \|\mathbf{D}_t\|_F = 1$ under their spectral-norm smoothness assumption, and remark that updating with $\bar{\mathbf{S}}_t$ directly is equivalent after absorbing $\sqrt{mn}$ into η_t. That equivalence holds for a single matrix but not across layers of differing shape, and their algorithm applies no shape factor, so their experiments use a single global rate as well. The substitution is also loose in a shape-dependent way: $\mathbf{s}$ has rank at most $\min(m,n)$, so $\|\mathbf{s}\|_F \le \sqrt{\min(m,n)}\,\|\mathbf{s}\|_{\text{op}}$ and the substitution of $\sqrt{mn}$ for the operator norm is loose by up to $\sqrt{\min(m,n)}$. That bound itself grows with depth, from $\sqrt{27} \approx 5.2$ at the stem of a ResNet-18 to $\sqrt{512} \approx 22.6$ in the last stage. The induced spectral trust region therefore drifts across the network instead of remaining flat.

A.18 Algorithms

Federated protocol. In the federated version of SignMuon, the server acts as an aggregation node, while the main optimization procedure is performed locally on the clients. At the beginning of communication round t, each client j receives a copy of the current global model and evaluates one stochastic gradient $\mathbf{G}_t^{(j)} = \nabla f_j(\mathbf{X}_t; \xi_t^{(j)})$ at it. Clients take *no* local parameter steps, so one round is one server step and no client-drift term arises. The released commands spell the batch as `--n_steps 3 --batch_size 64`, accumulating three mini-batches at fixed weights to save activation memory; since the BatchNorm statistics are frozen (Appendix A.10) the loss is separable across samples, so that average is identically a gradient at batch 192, and we report it as such.

In accordance with the general SignA framework (7), each client maintains its own local momentum buffer $\mathbf{M}_t^{(j)}$. After computing the gradient, the client updates the momentum estimate, applies the LMO (Newton–Schulz orthogonalization, Algorithm A1), and performs element-wise sign compression:

$$\mathbf{M}_t^{(j)} = \mu \mathbf{M}_{t-1}^{(j)} + (1-\mu)\,\mathbf{G}_t^{(j)}, \qquad \mathbf{D}_t^{(j)} = -A\big(\mathbf{M}_t^{(j)}\big), \qquad \mathbf{s}_t^{(j)} = \operatorname{sign}\big(\mathbf{D}_t^{(j)}\big),$$

where $t \in [0, \text{rounds} - 1]$. The momentum is written in exponential-moving-average form, matching Algorithms A8–A9 and the framework of Appendix A.9; it differs from the heavy-ball form of (7) by the positive factor $1/(1-\mu)$, which $\operatorname{sign}(\cdot)$ and the LMO absorb. Each client thus transmits one bit per matrix parameter component. On the server, the received signs are aggregated by majority vote, $\mathbf{s}_t^{\text{agg}} = \operatorname{sign}\big(\sum_{j=1}^N \mathbf{s}_t^{(j)}\big)$, which is ± 1 in each component: client messages are ± 1 valued by the

Algorithm A1: MuonLMO

Input: tensor $\mathbf{Y}$; Newton-Schulz coefficients $a = 3.4445,\ b = 4.7750,\ c = 2.0315$; iteration count $\mathrm{ns\_steps}$
Output: Muon LMO direction $\mathbf{D} \approx \mathbf{U}\mathbf{V}^\top$ of $\mathbf{Y}$

$\mathbf{Y} \leftarrow \mathrm{ReshapeTo2D}(\mathbf{Y})$ ▷ Flatten tensor to matrix $m \times n$
$\mathbf{Y} \leftarrow \mathbf{Y}/\|\mathbf{Y}\|_F$ ▷ Normalize (optional)
for $k = 1$ to $\mathrm{ns\_steps}$ **do** ▷ 5th-order Newton-Schulz orthogonalization
  $\mathbf{A} \leftarrow \mathbf{Y}\mathbf{Y}^\top$
  $\mathbf{Y} \leftarrow a\,\mathbf{Y} - b\,\mathbf{A}\mathbf{Y} + c\,\mathbf{A}^2\,\mathbf{Y}$
end for
$\mathbf{D} \leftarrow \mathrm{ReshapeToOriginal}(\mathbf{Y})$ ▷ Reshape back to the original tensor shape
return $\mathbf{D}$

convention of Section 4, so at an odd client count the vote cannot tie (at an even count a tie is broken by a fair coin). Since momentum has already been applied at the clients, the server directly updates the matrix parameters, $\mathbf{X}_{t+1} = \mathbf{X}_t - \eta_t \mathbf{s}_t^{\mathrm{agg}}$. The final classification layer is exempt from the SignMuon rule and optimized with AdamW.

Federated error feedback. Although federated SignMuon demonstrates strong empirical stability, sign-based compression inevitably introduces bias into the descent direction. To compensate for the quantization error and restore convergence guarantees, the error-feedback mechanism is adapted from EF21-Muon (Gruntkowska et al. 2025). The LMO orthogonalization is moved from the clients to the server: each client evaluates the difference between its local momentum $\mathbf{M}_t^{(j)}$ and the previous gradient estimate $\mathbf{g}_t^{(j)}$, compresses it via scaled sign, and sends only $(\mathbf{s}_t^{(j)}, \alpha_t^{(j)})$. The server aggregates these, updates a global gradient estimator $\mathbf{G}_{t+1}$, and applies the Muon LMO (Algorithm A9). The transmission of one additional scalar $\alpha_t^{(j)}$ per layer is insignificant overhead, keeping the uplink at ≈ 1 bit per parameter, with the full model broadcast on the downlink. All federated variants (the baseline majority-vote SignMuon and the error-feedback EF21-MuonUSign and EF21-MuonSign) thus preserve the geometric benefits of Muon-style matrix orthogonalization while maintaining an extremely low communication budget.

Algorithm A2: SignMuon

Input: Initial model $\mathbf{X}_0$, momentum coefficient μ, learning rate η_t
Output: Updated model $\mathbf{X}$

$\mathbf{M}_0 \leftarrow 0$
for $t = 1$ to T **do**
  Stochastic gradient $\mathbf{G}_t$
  $\mathbf{M}_t \leftarrow \mu \mathbf{M}_{t-1} + (1-\mu)\,\mathbf{G}_t$ ▷ Momentum accumulation
  $\tilde{\mathbf{M}}_t = \begin{cases} \mathbf{M}_t, & \text{(default)}, \\ (1-\mu)\,\mathbf{G}_t + \mu \mathbf{M}_t, & \text{(Nesterov)} \end{cases}$
  $\mathbf{D}_t \leftarrow \mathrm{MuonLMO}(\tilde{\mathbf{M}}_t)$
  $\mathbf{s}_t^{\uparrow} \leftarrow \mathrm{sign}(\mathbf{D}_t)$ ▷ Uplink sign compression
  $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t \mathbf{s}_t^{\uparrow}$ ▷ Update parameters
end for

All six federated methods are instances of just two templates, separated by *where the Muon LMO is evaluated*. When the sign acts *after* the LMO (the SignMuon family), each client must orthogonalize locally, so the LMO runs on the worker and the client transmits a compressed *direction* (Algorithm A8). When the sign acts *before* the LMO (the MuonUSign/MuonSign family), the client transmits a compressed *gradient*, and the server reconstructs it and applies a single LMO (Algorithm A9). Within each template, a method is fixed by its uplink compressor $\mathcal{C}^{\uparrow} \in \{\mathrm{sign}, \mathrm{EF21}\}$ and downlink compressor $\mathcal{C}^{\downarrow} \in \{\mathrm{exact}, \mathrm{sign}, \mathrm{EF21\text{-}P}\}$; Table A12 lists the six instantiations. As in the centralized setting, both templates are applied per matrix parameter, while vector parameters and the final classification layer are optimized with AdamW.

Algorithm A3: EF21-SignMuon

Input: Initial model $\mathbf{X}_0$, momentum coefficient μ, learning rate η_t
Output: Updated model $\mathbf{X}$

$\mathbf{M}_0 \leftarrow 0, \quad \mathbf{d}_0^{\text{est}} \leftarrow 0$
for $t = 1$ to T **do**
  Stochastic gradient $\mathbf{G}_t$
  $\mathbf{M}_t \leftarrow \mu \mathbf{M}_{t-1} + (1-\mu)\,\mathbf{G}_t$ ▷ Momentum accumulation (EMA)
  $\tilde{\mathbf{M}}_t = \begin{cases} \mathbf{M}_t, & \text{(default)}, \\ (1-\mu)\,\mathbf{G}_t + \mu \mathbf{M}_t, & \text{(Nesterov)} \end{cases}$
  $\mathbf{D}_t \leftarrow \mathrm{MuonLMO}(\tilde{\mathbf{M}}_t)$
  $\Delta_t^{\uparrow} \leftarrow \mathbf{D}_t - \mathbf{d}_{t-1}^{\text{est}}$ ▷ Uplink residual (LMO direction)
  $\alpha_t^{\uparrow} \leftarrow \mathrm{mean}(|\Delta_t^{\uparrow}|)$
  $\mathbf{d}_t^{\text{est}} \leftarrow \mathbf{d}_{t-1}^{\text{est}} + \alpha_t^{\uparrow}\,\mathrm{sign}(\Delta_t^{\uparrow})$ ▷ Uplink EF21
  $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t \mathbf{d}_t^{\text{est}}$ ▷ Update parameters
end for

Algorithm A4: MuonUSign

Input: Initial model $\mathbf{X}_0$, momentum coefficient μ, learning rate η_t
Output: Updated model $\mathbf{X}$

$\mathbf{M}_0 \leftarrow 0$
for $t = 1$ to T **do**
  Stochastic gradient $\mathbf{G}_t$
  $\mathbf{M}_t \leftarrow \mu \mathbf{M}_{t-1} + (1-\mu)\,\mathbf{G}_t$ ▷ Momentum accumulation
  $\tilde{\mathbf{M}}_t = \begin{cases} \mathbf{M}_t, & \text{(default)}, \\ (1-\mu)\,\mathbf{G}_t + \mu \mathbf{M}_t, & \text{(Nesterov)} \end{cases}$
  $\mathbf{s}_t^{\uparrow} \leftarrow \mathrm{sign}(\tilde{\mathbf{M}}_t)$ ▷ Uplink sign compression
  $\mathbf{D}_t \leftarrow \mathrm{MuonLMO}(\mathbf{s}_t^{\uparrow})$
  $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t \mathbf{D}_t$ ▷ Update parameters
end for

Algorithm A5: MuonSign

Input: Initial model $\mathbf{X}_0$, momentum coefficient μ, learning rate η_t
Output: Updated model $\mathbf{X}$

$\mathbf{M}_0 \leftarrow 0$
for $t = 1$ to T **do**
  Stochastic gradient $\mathbf{G}_t$
  $\mathbf{M}_t \leftarrow \mu \mathbf{M}_{t-1} + (1-\mu)\,\mathbf{G}_t$ ▷ Momentum accumulation
  $\tilde{\mathbf{M}}_t = \begin{cases} \mathbf{M}_t, & \text{(default)}, \\ (1-\mu)\,\mathbf{G}_t + \mu \mathbf{M}_t, & \text{(Nesterov)} \end{cases}$
  $\mathbf{s}_t^{\uparrow} \leftarrow \mathrm{sign}(\tilde{\mathbf{M}}_t)$ ▷ Uplink sign compression
  $\mathbf{D}_t \leftarrow \mathrm{MuonLMO}(\mathbf{s}_t^{\uparrow})$
  $\mathbf{s}_t^{\downarrow} \leftarrow \mathrm{sign}(\mathbf{D}_t)$ ▷ Downlink sign compression
  $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t \mathbf{s}_t^{\downarrow}$ ▷ Update parameters
end for

Algorithm A6: EF21-MuonUSign

Input: Initial model $\mathbf{X}_0$, momentum coefficient μ, learning rate η_t
Output: Updated model $\mathbf{X}$

$\mathbf{M}_0 \leftarrow 0, \quad \mathbf{g}_0^{\text{est}} \leftarrow 0$
for $t = 1$ to T **do**
  Stochastic gradient $\mathbf{G}_t$
  $\mathbf{M}_t \leftarrow \mu \mathbf{M}_{t-1} + (1-\mu)\,\mathbf{G}_t$ ▷ Momentum accumulation
  $\tilde{\mathbf{M}}_t = \begin{cases} \mathbf{M}_t, & \text{(default)}, \\ (1-\mu)\,\mathbf{G}_t + \mu \mathbf{M}_t, & \text{(Nesterov)} \end{cases}$
  $\Delta_t^{\uparrow} \leftarrow \tilde{\mathbf{M}}_t - \mathbf{g}_{t-1}^{\text{est}}$ ▷ Uplink residual
  $\alpha_t^{\uparrow} \leftarrow \operatorname{mean}(|\Delta_t^{\uparrow}|)$
  $\mathbf{g}_t^{\text{est}} \leftarrow \mathbf{g}_{t-1}^{\text{est}} + \alpha_t^{\uparrow} \operatorname{sign}(\Delta_t^{\uparrow})$ ▷ Uplink EF21
  $\mathbf{D}_t \leftarrow \operatorname{MuonLMO}(\mathbf{g}_t^{\text{est}})$
  $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t \mathbf{D}_t$ ▷ Update parameters
end for

Algorithm A7: EF21-MuonSign

Input: Initial model $\mathbf{X}_0 = \mathbf{W}_0$, momentum coefficient μ, learning rate η_t
Output: Updated model $\mathbf{X}$

$\mathbf{M}_0 \leftarrow 0, \quad \mathbf{g}_0^{\text{est}} \leftarrow 0$
for $t = 1$ to T **do**
  Stochastic gradient $\mathbf{G}_t$ at the broadcast model $\mathbf{W}_{t-1}$
  $\mathbf{M}_t \leftarrow \mu \mathbf{M}_{t-1} + (1-\mu)\,\mathbf{G}_t$ ▷ Momentum accumulation
  $\tilde{\mathbf{M}}_t = \begin{cases} \mathbf{M}_t, & \text{(default)}, \\ (1-\mu)\,\mathbf{G}_t + \mu \mathbf{M}_t, & \text{(Nesterov)} \end{cases}$
  $\Delta_t^{\uparrow} \leftarrow \tilde{\mathbf{M}}_t - \mathbf{g}_{t-1}^{\text{est}}$ ▷ Uplink residual
  $\alpha_t^{\uparrow} \leftarrow \operatorname{mean}(|\Delta_t^{\uparrow}|)$
  $\mathbf{g}_t^{\text{est}} \leftarrow \mathbf{g}_{t-1}^{\text{est}} + \alpha_t^{\uparrow} \operatorname{sign}(\Delta_t^{\uparrow})$ ▷ Uplink EF21
  $\mathbf{D}_t \leftarrow \operatorname{MuonLMO}(\mathbf{g}_t^{\text{est}})$
  $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t \mathbf{D}_t$ ▷ Update parameters (server)
  $\Delta_t^{\downarrow} \leftarrow \mathbf{X}_t - \mathbf{W}_{t-1}$ ▷ Downlink residual
  $\alpha_t^{\downarrow} \leftarrow \operatorname{mean}(|\Delta_t^{\downarrow}|)$
  $\mathbf{W}_t \leftarrow \mathbf{W}_{t-1} + \alpha_t^{\downarrow} \operatorname{sign}(\Delta_t^{\downarrow})$ ▷ Downlink EF21-P
end for

Method	**LMO**	**Uplink** $\mathcal{C}^{\uparrow}$	**Downlink** $\mathcal{C}^{\downarrow}$
SignMuon	worker	sign / MV	exact
EF21-SignMuon	worker	EF21	exact
MuonUSign	server	sign / MV	exact
MuonSign	server	sign / MV	sign
EF21-MuonUSign	server	EF21	exact
EF21-MuonSign	server	EF21	EF21-P

Table A12: The six federated methods as instantiations of the two templates: Algorithm A8 (worker-side LMO; rows 1–2) and Algorithm A9 (server-side LMO; rows 3–6). Each method is fixed by the LMO location and the uplink/downlink compressors. "MV": majority vote; "EF21-P": primal (model-side) error feedback; "exact": full-precision broadcast.

Algorithm A8: Federated SignMuon / EF21-SignMuon (worker-side LMO)

Input: initial model $\mathbf{X}_0$; clients N; rounds T; learning rate η_t; momentum μ; uplink compressor $\mathcal{C}^{\uparrow} \in \{\text{sign}, \text{EF21}\}$ (Table A12)
Output: global model $\mathbf{X}_T$

$\mathbf{M}_0^{(j)} \leftarrow 0$, $\mathbf{d}_0^{(j)} \leftarrow 0$ for all j; $\quad \mathbf{d}_0 \leftarrow 0$
for $t = 1$ to T **do**
    broadcast $\mathbf{X}_{t-1}$ to all clients ▷ exact downlink
    for $j = 1$ to N **in parallel do** ▷ client j
        $\mathbf{G}_t^{(j)} \leftarrow \nabla f_j(\mathbf{X}_{t-1}; \xi_t^{(j)})$
        $\mathbf{M}_t^{(j)} \leftarrow \mu \mathbf{M}_{t-1}^{(j)} + (1-\mu)\, \mathbf{G}_t^{(j)}$
        $\tilde{\mathbf{M}}_t^{(j)} \leftarrow \mathbf{M}_t^{(j)}$ ▷ or $(1-\mu)\mathbf{G}_t^{(j)} + \mu \mathbf{M}_t^{(j)}$ (Nesterov)
        $\mathbf{D}_t^{(j)} \leftarrow \mathrm{MuonLMO}\big(\tilde{\mathbf{M}}_t^{(j)}\big)$ ▷ LMO on the client
        if $\mathcal{C}^{\uparrow} =$ EF21 **then** ▷ EF21-SignMuon
            $\Delta_t^{(j)} \leftarrow \mathbf{D}_t^{(j)} - \mathbf{d}_{t-1}^{(j)}; \quad \alpha_t^{(j)} \leftarrow \mathrm{mean}(|\Delta_t^{(j)}|)$
            $\mathbf{d}_t^{(j)} \leftarrow \mathbf{d}_{t-1}^{(j)} + \alpha_t^{(j)}\, \mathrm{sign}(\Delta_t^{(j)})$
            send $\big(\mathrm{sign}(\Delta_t^{(j)}),\, \alpha_t^{(j)}\big)$
        else ▷ SignMuon
            send $\mathbf{s}_t^{(j)} \leftarrow \mathrm{sign}(\mathbf{D}_t^{(j)})$
        end if
    end for
    on the server:
    if $\mathcal{C}^{\uparrow} =$ EF21 **then**
        $\mathbf{d}_t \leftarrow \mathbf{d}_{t-1} + \frac{1}{N} \sum_j \alpha_t^{(j)}\, \mathrm{sign}(\Delta_t^{(j)})$ ▷ aggregate direction
        $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t\, \mathbf{d}_t$
    else
        $\hat{\mathbf{s}}_t \leftarrow \mathrm{sign}\big(\sum_j \mathbf{s}_t^{(j)}\big)$ ▷ majority vote
        $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t\, \hat{\mathbf{s}}_t$
    end if
end for

Algorithm A9: Federated MuonUSign / MuonSign / EF21-MuonUSign / EF21-MuonSign (server-side LMO)

Input: initial model $\mathbf{X}_0$ (and $\mathbf{W}_0 = \mathbf{X}_0$ if $\mathcal{C}^{\downarrow} =$ EF21-P); clients N; rounds T; learning rate η_t; momentum μ; compressors $\mathcal{C}^{\uparrow} \in \{\text{sign}, \text{EF21}\}$, $\mathcal{C}^{\downarrow} \in \{\text{exact}, \text{sign}, \text{EF21-P}\}$ (Table A12)
Output: global model $\mathbf{X}_T$

$\mathbf{M}_0^{(j)} \leftarrow 0$, $\mathbf{g}_0^{(j)} \leftarrow 0$ for all j; $\quad \mathbf{G}_0 \leftarrow 0$
broadcast $\mathbf{X}_0$ *once*; each client keeps a local copy $\mathbf{V}_0 \leftarrow \mathbf{X}_0$, refreshed below from the downlink message alone; $\mathbf{W}_0 \leftarrow \mathbf{X}_0$ if $\mathcal{C}^{\downarrow} =$ EF21-P
for $t = 1$ to T **do**
  for $j = 1$ to N **in parallel do** ▷ client j, holding $\mathbf{V}_{t-1}$
    $\mathbf{G}_t^{(j)} \leftarrow \nabla f_j(\mathbf{V}_{t-1}; \xi_t^{(j)})$
    $\mathbf{M}_t^{(j)} \leftarrow \mu \mathbf{M}_{t-1}^{(j)} + (1-\mu)\,\mathbf{G}_t^{(j)}$
    $\tilde{\mathbf{M}}_t^{(j)} \leftarrow \mathbf{M}_t^{(j)}$ ▷ or $(1-\mu)\mathbf{G}_t^{(j)} + \mu \mathbf{M}_t^{(j)}$ (Nesterov)
    if $\mathcal{C}^{\uparrow} =$ EF21 **then** ▷ EF21-MuonUSign / EF21-MuonSign
      $\Delta_t^{(j)} \leftarrow \tilde{\mathbf{M}}_t^{(j)} - \mathbf{g}_{t-1}^{(j)}$; $\quad \alpha_t^{(j)} \leftarrow \mathrm{mean}(|\Delta_t^{(j)}|)$
      $\mathbf{g}_t^{(j)} \leftarrow \mathbf{g}_{t-1}^{(j)} + \alpha_t^{(j)}\,\mathrm{sign}(\Delta_t^{(j)})$
      send $\big(\mathrm{sign}(\Delta_t^{(j)}),\, \alpha_t^{(j)}\big)$
    else ▷ MuonUSign / MuonSign
      send $\mathbf{s}_t^{(j)} \leftarrow \mathrm{sign}\big(\tilde{\mathbf{M}}_t^{(j)}\big)$
    end if
  end for
  on the server:
  if $\mathcal{C}^{\uparrow} =$ EF21 **then**
    $\mathbf{G}_t \leftarrow \mathbf{G}_{t-1} + \frac{1}{N}\sum_j \alpha_t^{(j)}\,\mathrm{sign}(\Delta_t^{(j)})$ ▷ reconstruct gradient
  else
    $\mathbf{G}_t \leftarrow \mathrm{sign}\big(\sum_j \mathbf{s}_t^{(j)}\big)$ ▷ majority vote
  end if
  $\mathbf{D}_t \leftarrow \mathrm{MuonLMO}(\mathbf{G}_t)$ ▷ single LMO on the server
  if $\mathcal{C}^{\downarrow} =$ sign **then** ▷ MuonSign: 1 bit/param down
    $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t\,\mathrm{sign}(\mathbf{D}_t)$; **broadcast** $\mathrm{sign}(\mathbf{D}_t)$
    each client: $\mathbf{V}_t \leftarrow \mathbf{V}_{t-1} - \eta_t\,\mathrm{sign}(\mathbf{D}_t)$ ▷ $= \mathbf{X}_t$; one model
  else if $\mathcal{C}^{\downarrow} =$ EF21-P **then** ▷ EF21-MuonSign: 1 bit/param down
    $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t \mathbf{D}_t$ ▷ server model
    $\Delta_t^{\downarrow} \leftarrow \mathbf{X}_t - \mathbf{W}_{t-1}$; $\quad \alpha_t^{\downarrow} \leftarrow \mathrm{mean}(|\Delta_t^{\downarrow}|)$
    $\mathbf{W}_t \leftarrow \mathbf{W}_{t-1} + \alpha_t^{\downarrow}\,\mathrm{sign}(\Delta_t^{\downarrow})$; **broadcast** $\big(\mathrm{sign}(\Delta_t^{\downarrow}), \alpha_t^{\downarrow}\big)$
    each client: $\mathbf{V}_t \leftarrow \mathbf{V}_{t-1} + \alpha_t^{\downarrow}\,\mathrm{sign}(\Delta_t^{\downarrow})$ ▷ $= \mathbf{W}_t$, broadcast model
  else ▷ exact downlink: MuonUSign / EF21-MuonUSign
    $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t \mathbf{D}_t$; **broadcast** $\mathbf{X}_t$; each client: $\mathbf{V}_t \leftarrow \mathbf{X}_t$
  end if
end for

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